Modular Forms

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1 Congruence Subgroups and Modular Curves

For any complex z let

$$e(z) = e^{2\pi i z}.$$

Also, we will distinguish the primitive third root of unity $\omega = e\left(\frac{1}{3}\right)$. All sums and products over a quotient group will be consider to be over a complete set of representatives and we identify cosets with corresponding representatives. If an integer a is taken modulo m, for any positive integer m, we write $\overline{a}$ for the inverse of a.

1.1 Congruence Subgroups

To define congruence subgroups, we first need to introduce an associated group. We call $\mathrm{SL}_2(\mathbb{Z})$ the **modular group** which is the group of matrices with integer entries and of determinant 1. We will distinguish two matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The first matrix is of order 4 while the second has infinite order. The first result usually proved about the modular group is that it is generated by S and T.

Proposition 1.1.

$$\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle.$$

Proof. Let n be an integer and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. We will prove $\gamma \in \langle S, T \rangle$ by demonstrating that it can be reduced to the identity matrix. If $c = 0$ then γ is the identity matrix since $\det(\gamma) = 1$ and we are done. So suppose $c \neq 0$ and write $a = qc + r$ for some integers q and r with $|r| < |c|$. A short computation shows

$$ST^{-q}\gamma = \begin{pmatrix} -c & -d \\ r & b - qd \end{pmatrix}.$$

The upper-left entry of this matrix strictly larger than the lower-left entry in absolute value. Therefore if we repeatedly apply this argument, it must terminate with the lower-left entry vanishing. But then we have reached the identity matrix. □

We can now introduce congruence subgroups which are special subgroups of the modular group defined by congruence conditions on their entries. We begin with principal congruence subgroups. For a positive integer N, the **principal congruence subgroup** $\Gamma(N)$ of level N is defined by

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

This congruence subgroup consists of the matrices which reduce to the identity modulo N. Whence $\Gamma(N)$ is the kernel of the homomorphism

$$\pi : \mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \pmod{N},$$

and is therefore a normal subgroup. A subgroup Γ of the modular group is said to be a **congruence subgroup** if it contains a principal congruence subgroup. The minimal such N for which $\Gamma(N) \subseteq \Gamma$ is called the **level** of Γ. Note that if $M \mid N$ then $\Gamma(N) \subseteq \Gamma(M)$. Thus if Γ is a congruence subgroup of level N then it contains the principal congruence subgroups $\Gamma(kN)$ for all positive integers k. This implies that congruence subgroups are preserved under intersection. In fact, if Γ_1 and Γ_2 are congruence subgroups of levels N_1 and N_2 then $\Gamma_1 \cap \Gamma_2$ is guaranteed to be a congruence subgroup of level at most $[N_1, N_2]$. Also, it turns out that the aforementioned homomorphism is surjective.

Proposition 1.2. *Let N be a positive integer. Then the homomorphism*

$$\pi : \mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \pmod{N},$$

given by natural projection, is surjective.

Proof. Let U and L be the matrices in $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ given by

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

A short computation shows that these are the images of T and TST under π respectively. So Proposition 1.1 will imply that π is surjective provided U and L generate $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. So let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. We will prove $\gamma \in \langle U, L \rangle$ by demonstrating that it can be reduced to the identity matrix. To show this, observe that $U^n = U(n)$ and $L^n = L(n)$ where $U(n)$ and $L(n)$ are the upper and lower unipotent matrices

$$U(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad L(n) = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix},$$

for n modulo N. So it suffices to show that this collection of unipotent matrices generates $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$. If $c = 0$ then $\det(\gamma) = 1$ implies that $d = a^{-1}$ so that $\gamma = \left(\begin{smallmatrix} a & b \\ 0 & a^{-1} \end{smallmatrix}\right)$. Short computations show

$$U(-ab)\gamma = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix},$$

and

$$L(a^{-1}-1)U(1)L(a-1)U(-a^{-1}) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

Whence γ can be reduced to the identity matrix. If $c \neq 0$ then $\det(\gamma) = 1$ and therefore c is invertible. A short computation shows

$$L(-c)U((1-a)c^{-1})\gamma = \begin{pmatrix} 1 & (d-1)c^{-1} \\ 0 & 1 \end{pmatrix},$$

and we recognize this latter matrix is as $U((d-1)c^{-1})$. Whence γ can again be reduced to the identity matrix. $\square$

As $\Gamma(N)$ is the kernel of this homomorphism, the result implies $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = |\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})|$. Since any congruence subgroup Γ contains a principal congruence subgroup of some level, it follows that Γ has finite index in $\mathrm{SL}_2(\mathbb{Z})$. Accordingly, we set

$$d_\Gamma = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma].$$

We now distinguish two important classes of congruence subgroups other than the principal ones. These are the congruence subgroups

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod N \right\},$$

and

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod N \right\},$$

both of which are of level N. These congruence subgroups consist of the matrices which reduce to unipotent upper triangular and upper triangular modulo N respectively. The latter subgroup $\Gamma_0(N)$ is called the **Hecke congruence subgroup** of level N. Note that $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N)$.

Since the principal congruence subgroups are normal in $\mathrm{SL}_2(\mathbb{Z})$, if Γ is a congruence subgroup of level N then so is $\alpha^{-1}\Gamma\alpha$ for any $\alpha \in \mathrm{SL}_2(\mathbb{Z})$. In fact, something more general holds. Congruence subgroups are preserved under conjugation by elements of $\mathrm{GL}_2^+(\mathbb{Q})$ provided we restrict to those elements in $\mathrm{SL}_2(\mathbb{Z})$.

Lemma 1.3. *Let Γ be a congruence subgroup and $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then $\alpha^{-1}\Gamma\alpha \cap \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup. In particular, if $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ then $\alpha^{-1}\Gamma\alpha$ is a congruence subgroup.*

Proof. Let the level of Γ be N and choose a positive integer r such that $r\alpha, r\alpha^{-1} \in \mathrm{GL}_2^+(\mathbb{Z})$. Let $M = r^2N$ and notice that any $\gamma \in \Gamma(M)$ is of the form

$$\gamma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + M \begin{pmatrix} k_1 & k_2 \\ k_3 & k_4 \end{pmatrix},$$

for integers $k_1, \dots, k_4$. Therefore $\Gamma(M) \subseteq I + M\mathrm{Mat}_2(\mathbb{Z})$ and so

$$\alpha\Gamma(M)\alpha^{-1} \subseteq I + N\mathrm{Mat}_2(\mathbb{Z}).$$

As $\Gamma(N) = (I + N\mathrm{Mat}_2(\mathbb{Z})) \cap \mathrm{SL}_2(\mathbb{Z})$, it follows that $\alpha\Gamma(M)\alpha^{-1} \subseteq \Gamma(N)$. Whence $\Gamma(M) \subseteq \alpha^{-1}\Gamma\alpha$ because Γ is of level N. Intersecting with $\mathrm{SL}_2(\mathbb{Z})$ proves the first statement. The second statement is immediate. $\square$

Note that in the case $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$, normality of principal congruence subgroups force Γ and $\alpha^{-1}\Gamma\alpha$ to be of the same level. In particular, this will always happen whenever we conjugate by elements of the modular group.

Also, since congruence subgroups are preserved under intersection, the previous lemma further implies that $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ is a congruence subgroup for any two congruence subgroups Γ_1 and Γ_2 and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. In this setting, consider the double coset

$$\Gamma_1\alpha\Gamma_2 = \{\gamma_1\alpha\gamma_2 : \gamma_1 \in \Gamma_1 \text{ and } \gamma_2 \in \Gamma_2\}.$$

Note that Γ_1 acts on $\Gamma_1\alpha\Gamma_2$ by left multiplication so that it decomposes into a disjoint union of orbits. That is,

$$\Gamma_1\alpha\Gamma_2 = \bigcup_{\beta \in \Gamma_1\backslash\Gamma_1\alpha\Gamma_2} \Gamma_1\beta.$$

It turns out that this union is finite. To see this, we need to relate the orbit space $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ to a quotient of Γ_2 with respect to the action of some subgroup. Consider the congruence subgroup $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ which is a subgroup of Γ_2. As this subgroup acts on Γ_2 by left multiplication, we again have an orbit space $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$. We first show how these orbit spaces are related.

Lemma 1.4. *Let Γ_1 and Γ_2 be congruence subgroups and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then the right multiplication map*

$$\Gamma_2 \to \Gamma_1\backslash\Gamma_1\alpha\Gamma_2 \qquad \gamma_2 \mapsto \Gamma_1\alpha\gamma_2,$$

induces a bijection from $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$ to $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$.

Proof. We first show that this map is $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)$-invariant so that it induces a map on the quotient. Indeed, if $\gamma \in \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ then $\gamma = \alpha^{-1}\gamma_1\alpha$ for some $\gamma_1 \in \Gamma_1$. Thus $\Gamma_1\alpha\gamma\gamma_2 = \Gamma_1\alpha\gamma_2$ as claimed. We now show that the induced map is a bijection.

We will prove injectivity and then surjectivity. For the former, let γ_2 and γ_2' be representatives of orbits of $(\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2)\backslash\Gamma_2$ such that $\Gamma_1\alpha\gamma_2 = \Gamma_1\alpha\gamma_2'$. This implies $\alpha\gamma_2(\gamma_2')^{-1} \in \Gamma_1\alpha$ and so $\gamma_2(\gamma_2')^{-1} \in \alpha^{-1}\Gamma_1\alpha$. But we also have $\gamma_2(\gamma_2')^{-1} \in \Gamma_2$ and these two facts together imply that $\gamma_2(\gamma_2')^{-1} \in \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$. Hence

$$(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma_2 = (\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma_2',$$

from which injectivity follows. For the latter, any representative β of an orbit in $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ is of the form $\beta = \gamma_1\alpha\gamma_2$ for some $\gamma_1 \in \Gamma_1$ and $\gamma_2 \in \Gamma_2$. But then γ_2 represents an orbit of $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\backslash\Gamma_2$ mapping to the orbit represented by β as $\Gamma_1\beta = \Gamma_1\alpha\gamma_2$. This shows surjectivity. $\square$

With this lemma in hand, we can prove that the aforementioned disjoint union is finite.

Proposition 1.5. *Let Γ_1 and Γ_2 be congruence subgroups and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then the disjoint union*

$$\Gamma_1\alpha\Gamma_2 = \bigcup_{\beta\in\Gamma_1\backslash\Gamma_1\alpha\Gamma_2} \Gamma_1\beta,$$

is finite.

Proof. By Lemma 1.4, the number of orbits of $\Gamma_1\backslash\Gamma_1\alpha\Gamma_2$ is in bijection with those of $(\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2)\backslash\Gamma_2$ which is $[\Gamma_2 : \alpha^{-1}\Gamma_1\alpha\cap\Gamma_2]$. By Lemma 1.3, $\alpha^{-1}\Gamma_1\alpha\cap\mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup and thus $\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2$ is as well. It follows that $[\Gamma_2 : \alpha^{-1}\Gamma_1\alpha\cap\Gamma_2] \leq d_{\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2}$ where $d_{\alpha^{-1}\Gamma_1\alpha\cap\Gamma_2}$ is finite which completes the proof. $\square$

1.2 Modular Curves and Fundamental Domains

Recall that $\mathrm{GL}_2^+(\mathbb{R})$ acts on the Riemann sphere $\hat{\mathbb{C}}$ by Möbius transformations. Explicitly, any $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ acts on $z \in \hat{\mathbb{C}}$ by

$$\alpha z = \frac{az+b}{cz+d}.$$

Moreover, this group action is invariant under scalar multiplication and negation and acts by automorphisms of $\hat{\mathbb{C}}$. A short computation shows

$$\mathrm{Im}(\alpha z) = \det(\alpha)\frac{\mathrm{Im}(z)}{|cz+d|^2}, \tag{1}$$

Therefore α preserves the sign of the imaginary part of z and therefore preserves the upper half-plane $\mathbb{H}$, the lower half-plane $\overline{\mathbb{H}}$, and the extended real line $\hat{\mathbb{R}}$. Whence $\mathrm{GL}_2^+(\mathbb{R})$ acts by automorphisms on these subspaces respectively. In the case of $\mathrm{GL}_2^+(\mathbb{Q})$, we also have a restricted action on $\hat{\mathbb{Q}}$ by automorphisms since the entries of these matrices are rational. Moreover, if $\alpha \in \mathrm{SL}_2(\mathbb{R})$ then we have the simplified relation

$$\mathrm{Im}(\alpha z) = \frac{\mathrm{Im}(z)}{|cz+d|^2}. \tag{2}$$

An important restriction of this action is to the modular group $\mathrm{SL}_2(\mathbb{Z})$. Again, these matrices act on $\hat{\mathbb{C}}$ by Möbius transformations and preserves the aforementioned subspaces $\mathbb{H}$, $\overline{\mathbb{H}}$, $\hat{\mathbb{R}}$, and $\hat{\mathbb{Q}}$ respectively.

We will primarily be interested in certain actions of subgroups of $\mathrm{SL}_2(\mathbb{R})$ on $\mathbb{H}$. A **Fuchsian group** is any discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ that acts properly discontinuously on $\mathbb{H}$. The modular group is a prime example of a Fuchsian group.

Proposition 1.6. *The modular group is a Fuchsian group.*

Proof. $\mathrm{SL}_2(\mathbb{Z})$ is discrete in $\mathrm{SL}_2(\mathbb{R})$ because $\mathbb{Z}$ is discrete in $\mathbb{R}$. So it remains to show that $\mathrm{SL}_2(\mathbb{Z})$ acts properly discontinuously on $\mathbb{H}$. To prove this, it suffices to show that for any compact subset $K \subset \mathbb{H}$ there are finitely many $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ satisfying $\gamma K \cap K \neq 0$. Set $M = \min_{z \in K} \mathrm{Im}(z)$. If $\gamma K \cap K \neq \varnothing$ then there exists a $z \in K$ such that $\mathrm{Im}(\gamma z) \geq M$. This inequality is equivalent to

$$(cx+d)^2 + (cy)^2 \leq \frac{y}{M},$$

by Equation (2), which is only satisfied for finitely many pairs of relatively prime integers (c, d) since $y > 0$. Each pair determines infinitely many matrices $\gamma_k = \left(\begin{smallmatrix} a+kc & b+kd \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ possibly satisfying $\gamma_k K \cap K \neq \varnothing$. As K is compact it has bounded real part and so only finitely many such k satisfy this condition. Hence there are finitely many possible matrices γ. $\square$

As an immediate consequence of this result, any subgroup of the modular group is also Fuchsian. In particular, all congruence subgroups are Fuchsian. Fuchsian groups are important because proper discontinuity of the group action ensures that the corresponding quotient of $\mathbb{H}$ wil be connected Hausdorff because $\mathbb{H}$ is. The principal example of such quotients are modular curves.

A **modular curve** is a quotient $\Gamma\backslash\mathbb{H}$ of the upper half-plane $\mathbb{H}$ by the action of a congruence subgroup Γ. Then, as remarked before, $\Gamma\backslash\mathbb{H}$ is connected Hausdorff because Γ is Fuchsian. In particular, $\Gamma\backslash\mathbb{H}$ admits a fundamental domain

$$\mathcal{F} = \left\{ z \in \mathbb{H} : |\mathrm{Re}(z)| \leq \frac{1}{2} \text{ and } |z| \geq 1 \right\},$$

as the following proposition shows:

Proposition 1.7. *$\mathcal{F}$ is a fundamental domain for the modular group.*

Proof. We first show any point in $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to a point in $\mathcal{F}$. So let $z \in \mathbb{H}$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then Equation (2) gives

$$\mathrm{Im}(\gamma z) = \frac{y}{|cz+d|^2}.$$

Since $\det(\gamma) = 1$ we cannot have $c = d = 0$. As $y > 0$, it is easily checked that $|cz + d| \geq \min(y, 1)$ and thus bounded away from zero. Whence there are finitely many pairs of relatively prime integers (c, d) such that $|cz + d|^2 \leq M$ for any $M > 0$. So choose a γ such that it minimizes $|cz + d|$ and hence maximizes $\mathrm{Im}(\gamma z)$. In particular, $\mathrm{Im}(S\gamma z) \leq \mathrm{Im}(\gamma z)$ which is to say

$$\frac{\mathrm{Im}(\gamma z)}{|\gamma z|^2} \leq \mathrm{Im}(\gamma z).$$

The inequality implies $|\gamma z| \geq 1$. Since $\mathrm{Im}(T^n\gamma z) = \mathrm{Im}(\gamma z)$ for all integers n, repeating the argument above with $T^n\gamma$ in place of γ, we see that $|T^n\gamma z| \geq 1$. But T shifts the real part by 1 so we can choose n such that $|\mathrm{Re}(T^n\gamma z)| \leq \frac{1}{2}$. Therefore $T^n\gamma$ maps z into $\mathcal{F}$. It remains to show that no two distinct interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. So suppose to the contrary that there exist distinct $z, z' \in \mathcal{F}^\circ$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $z' = \gamma z$. Without loss of generality, we may assume $\mathrm{Im}(z') \geq \mathrm{Im}(z)$. Then

$$\mathrm{Im}(z') = \frac{\mathrm{Im}(z)}{|cz + d|^2},$$

by Equation (2) and so $|cz + d| \leq 1$. As $|z| > 1$ and $|\mathrm{Re}(z)| < \frac{1}{2}$, short computations show that this inequality is valid only if $c = 0$. Then $\det(\gamma) = 1$ forces $a = d = \pm 1$. Finally, $|\mathrm{Re}(z)| < \frac{1}{2}$ further forces $b = 0$. But then $\gamma = \pm I$ which gives a contradiction since we assumed z and z' were distinct. $\square$

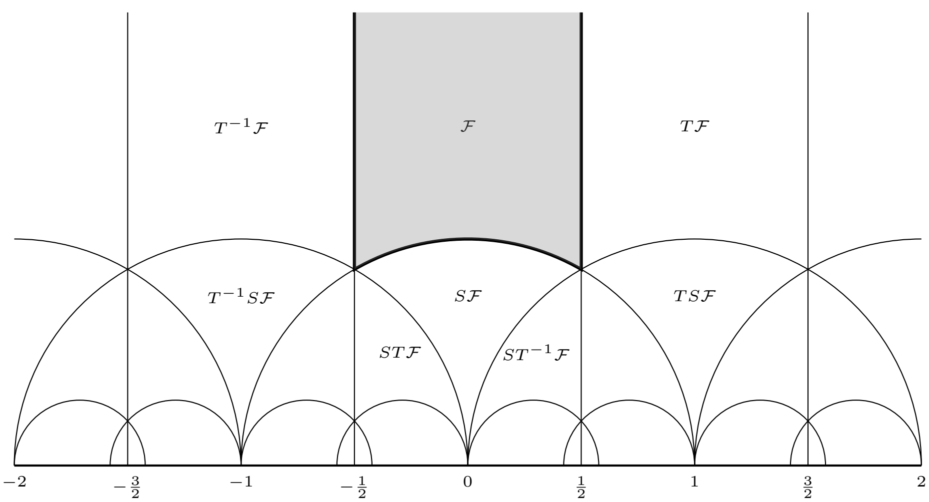

Figure 1: The standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$.

The region $\mathcal{F}$, shaded in Figure 1, is called the **standard fundamental domain**. This figure also displays how this fundamental domain changes under the actions of the generators of $\mathrm{SL}_2(\mathbb{Z})$ as in Proposition 1.1.

Remark 1.8. Any $z \in \mathcal{F}$ satisfies $\mathrm{Im}(z) \geq \frac{\sqrt{3}}{2}$. Indeed, any $z \in \mathcal{F}$ occurs on the arc of a circle of radius $r \geq 1$ subject to $|\mathrm{Re}(z)| \leq \frac{1}{2}$. Then $\mathrm{Im}(z)$ is minimized on the arc $|z| = 1$, and on this arc

$$\mathrm{Im}(z) = \sqrt{1 - \mathrm{Re}(z)^2}.$$

The right-hand side is minimized at $\mathrm{Re}(z) = \pm\frac{1}{2}$ giving $\mathrm{Im}(z) = \frac{\sqrt{3}}{2}$. This corresponds to the points $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $\frac{1}{2} + i\frac{\sqrt{3}}{2}$ which are ω and $1+\omega$ respectively. They are the vertices of the boundary of $\mathcal{F}$.

Any other modular curve $\Gamma\backslash\mathbb{H}$ admits a fundamental domain

$$\mathcal{F}_\Gamma = \bigcup_{\gamma \in \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} \gamma\mathcal{F},$$

obtained from the standard fundamental domain using representatives of orbits of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ as the following proposition shows:

Proposition 1.9. *Let Γ be a congruence subgroup. Then $\mathcal{F}_\Gamma$ is a fundamental domain for $\Gamma\backslash\mathbb{H}$.*

Proof. We first show that every point in $\mathbb{H}$ is Γ-equivalent to a point in $\mathcal{F}_\Gamma$. From the definition of $\mathcal{F}_\Gamma$, it is straightforward to check that

$$\bigcup_{\gamma' \in \Gamma} \gamma'\mathcal{F}_\Gamma = \mathbb{H}.$$

Whence every point in $\mathbb{H}$ is Γ-equivalent to a point in $\mathcal{F}_\Gamma$. We will now show that no two distinct interior points of $\mathcal{F}_\Gamma$ are Γ-equivalent. Suppose to the contrary that there are distinct $z_1, z_2 \in \mathcal{F}_\Gamma^\circ$ and a $\gamma \in \Gamma$ such that $\gamma z_1 = z_2$. In particular, $z_1 \in \gamma_1\mathcal{F}^\circ$ and $z_2 \in \gamma_2\mathcal{F}^\circ$ for some $\gamma_1, \gamma_2 \in \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$. As no two distinct interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent, this forces γ_1 and γ_2 to belong to the same coset of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$. Whence $\gamma_1 = \gamma_2$ and implying $z_1, z_2 \in \gamma_1\mathcal{F}^\circ \cap \gamma_2\mathcal{F}^\circ$. This gives a contradiction since no two distinct interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. Hence no two distinct interior points of $\mathcal{F}_\Gamma$ are Γ-equivalent. $\square$

It is also possible to say something about the stabilizer subgroups of points in $\mathbb{H}$ with respect to the action of a congruence subgroup Γ. Let Γ_z denote the stabilizer subgroup of $z \in \mathbb{H}$. Then $\Gamma_{\gamma z} = \gamma\Gamma_z\gamma^{-1}$ for any $\gamma \in \Gamma$ and it follows that all of the stabilizer subgroups are determined by those for $z \in \mathcal{F}_\Gamma$. Moreover, since $\mathcal{F}_\Gamma$ is a fundamental domain, Γ_z is trivial if $z \in \mathcal{F}_\Gamma^\circ$. So any point with a nontrivial stabilizer subgroups must lie on the boundary of $\mathcal{F}_\Gamma$. As Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$, Γ_z is a subgroup of $\mathrm{SL}_2(\mathbb{Z})_z$. This leads us to determining the stabilizer subgroups in the case of the modular group.

Proposition 1.10. *Let $z \in \mathcal{F}$. Then*

$$\mathrm{SL}_2(\mathbb{Z})_z = \begin{cases} \langle S \rangle & \text{if } z = i, \\ \langle ST \rangle & \text{if } z = \omega, \\ \langle TS \rangle & \text{if } z = 1+\omega, \\ \langle -I \rangle & \text{if } z \neq i, \omega, 1+\omega. \end{cases}$$

Moreover, $\mathrm{SL}_2(\mathbb{Z})_i$ *is of order* 4 *while* $\mathrm{SL}_2(\mathbb{Z})_\omega$ *and* $\mathrm{SL}_2(\mathbb{Z})_{1+\omega}$ *are of order* 6.

Proof. Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then $\gamma z = z$ implies

$$cz^2 + (d-a)z - b = 0.$$

We will analyze this identity on a case by case basis. In particular, we will take $z = i$, $z = \omega$, $z = 1+\omega$, and $z \neq i, \omega, 1+\omega$ respectively.

Taking $z = i$ in this identity gives

$$-c + (d-a)i - b = 0.$$

Equating the real and imaginary parts shows that $b = -c$ and $d = a$. Then $\gamma = \left(\begin{smallmatrix} a & -c \\ c & a \end{smallmatrix}\right)$ and $a^2 + c^2 = 1$. Therefore we have $a = 1$ and $c = 0$, $a = -1$ and $c = 0$, $a = 0$ and $c = 1$, or $a = 0$ and $c = -1$. These four cases correspond to the matrices I, $-I$, S, and $-S$ respectively. Since $S^2 = -I$, it follows that $\mathrm{SL}_2(\mathbb{Z})_i = \langle S \rangle$ and is of order 4.

Taking $z = \omega$ in the identity instead gives

$$c\omega^2 + (d-a)\omega - b = 0.$$

Recall that $m_\omega(x) = x^2 + x + 1$ is the minimal polynomial for ω over $\mathbb{Q}$. So $\omega^2 = -\omega - 1$ and substituting this identity into the one above yields

$$(d-a-c)\omega - (b+c) = 0.$$

By minimality of $m_\omega(x)$, ω and 1 are linearly independent over $\mathbb{Q}$ so we may equate coefficients to see that $d = a+c$ and $b = -c$. Then $\gamma = \left(\begin{smallmatrix} a & -c \\ c & a+c \end{smallmatrix}\right)$ and $a^2 + ac + c^2 = 1$. Completing the square shows $(a+c)^2 - ac = 1$. Therefore we have $a = 1$ and $c = 0$, $a = -1$ and $c = 0$, $a = 0$ and $c = 1$, $a = 0$ and $c = -1$, $a = -1$ and $c = 1$, or $a = 1$ and $c = -1$. These six cases correspond to the matrices I, $-I$, ST, $(ST)^4$, $(ST)^2$, and $(ST)^5$ respectively. As $(ST)^3 = -I$, it follows that $\mathrm{SL}_2(\mathbb{Z})_\omega = \langle ST \rangle$ and is of order 6. Moreover, if $z = 1+\omega$ then $z = T\omega$. Whence $\mathrm{SL}_2(\mathbb{Z})_{1+\omega} = T\mathrm{SL}_2(\mathbb{Z})_\omega T^{-1} = \langle TS \rangle$ and is of order 6.

It remains to show that if $z \neq i, \omega, 1+\omega$ then $\mathrm{SL}_2(\mathbb{Z})_z = \langle I \rangle$. Equivalently, will show the contrapositive which is to say that if $\mathrm{SL}_2(\mathbb{Z})_z \neq \langle -I \rangle$ then $z = i, \omega, 1+\omega$. Since the stabilizer subgroup of z is trivial if $z \in \mathcal{F}^\circ$, we may additionally assume z lies on the boundary of $\mathcal{F}$. Now suppose for such z that $\gamma \neq \pm I$. We first claim $c > 0$. For if $c = 0$, $\det(\gamma) = 1$ implies $a = d = \pm 1$. This further forces $b = 0$ implying $\gamma = \pm I$ which is a contradiction. Hence $c \neq 0$ and so z satisfies the aforementioned identity which is a quadratic. A short computation shows that the discriminant is $(a+d)^2 - 4$. Since this quadratic has integer coefficients and z is complex, the discriminant must be negative. So

$$(a+d)^2 - 4 < 0,$$

which forces $a + d = 0, 1, -1$. These three cases correspond to the discriminant being -4 in the first case and -3 in the latter two cases. Whence the quadratic formula implies

$$z = \frac{a \pm i}{c}, \quad \text{or} \quad z = \frac{2a - 1 \pm \sqrt{3}i}{2c}, \quad \text{or} \quad z = \frac{2a + 1 \pm \sqrt{3}i}{2c},$$

respectively. In any case, the bound $\mathrm{Im}(z) \geq \frac{\sqrt{3}}{2}$ by Remark 1.8 implies $c = \pm 1$. This forces $a = 0$ in the first case, $a = 0, 1$ in the second, and $a = 0, -1$ in the third. Whence $z = i$ in the first case and $z = \omega, 1+\omega$ in the second and third cases. This completes the proof. $\square$

Returning to the case of a congruence subgroup Γ, it follows from this result and our previous remarks that Γ_z is necessarily trivial if z is not $\mathrm{SL}_2(\mathbb{Z})$-equivalent to i, ω, or $1+\omega$. Moreover, if z is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to one of these points then Γ_z is trivial unless at least one of the matrices $\pm S$, $\pm ST$, or $\pm TS$ belongs to Γ respectively. In particular, Γ_z is always finite.

1.3 Cusps

Let Γ be a congruence subgroup. Notice that $\mathcal{F}_\Gamma$ is not compact as it doesn't contain any points of $\hat{\mathbb{Q}}$. However, if we consider $\mathcal{F}_\Gamma \cup \hat{\mathbb{Q}}$ as a subset of $\hat{\mathbb{C}}$ then it would appear that this space is compact. The elements of $\hat{\mathbb{Q}}$ correspond to cusps which we now define. Really that any $\gamma \in \Gamma$ acts on $\hat{\mathbb{Q}}$. A **cusp class** of $\Gamma\backslash\mathbb{H}$ is a coset of $\Gamma\backslash\hat{\mathbb{Q}}$. There is only one cusp class for the modular group since its action is transitive. Indeed, $\hat{\mathbb{Q}}$ is exactly the orbit of ∞. This is clear for ∞ and if $\frac{a}{c} \in \mathbb{Q}$ with $(a,c)=1$ then Bézout's identity provides the existence of integers b and d such that $ad-bc=1$ whence $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ satisfies $\gamma\infty = \frac{a}{c}$. Therefore the action is transitive. As Γ has finite index in the modular group and $\hat{\mathbb{Q}} = \mathrm{SL}_2(\mathbb{Z})\infty$, there can only be finitely many cosets of $\Gamma\backslash\hat{\mathbb{Q}}$ and the number of such is at most d_Γ. In other words, $\Gamma\backslash\mathbb{H}$ has at most d_Γ cusp classes. Also, the orbit of ∞ is a cusp class of $\Gamma\backslash\mathbb{H}$. A **cusp** is a representative of a cusp class. We denote cusps by gothic characters $\mathfrak{a}$, $\mathfrak{b}$, etc.

Remark 1.11. The cusps are exactly the points needed to make a fundamental domain $\mathcal{F}_\Gamma$ compact as a subset of $\hat{\mathbb{C}}$. To see this, suppose $\mathfrak{a}$ is a limit point of $\mathcal{F}_\Gamma$ that does not belong to $\mathcal{F}_\Gamma$. Then $\mathfrak{a} \in \hat{\mathbb{R}}$. As the unique limit point of $\mathcal{F}$ not belonging to $\mathcal{F}$ is ∞ and $\hat{\mathbb{Q}} = \mathrm{SL}_2(\mathbb{Z})\infty$, Proposition 1.9 implies $\mathfrak{a} \in \hat{\mathbb{Q}}$ which means $\mathfrak{a}$ is a cusp. In the case of the standard fundamental domain $\mathcal{F}$, the only cusp is ∞.

We will now describe the stabilizer subgroups of Γ relative to a cusp $\mathfrak{a}$. This will slightly depend upon if Γ contains $-I$, so let $\varepsilon_\Gamma = 1, 0$ according to if $-I \in \Gamma$ or not. We will denote the stabilizer subgroup of the cusp $\mathfrak{a}$ by $\Gamma_\mathfrak{a}$. We begin by describing Γ_∞. For if $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$ stabilizes ∞ then necessarily $c=0$ and since $\det(\gamma)=1$ we must have $a=d=\pm 1$ or $a=d=1$ according to if $-I \in \Gamma$ or not. In either case, γ acts on $\mathbb{H}$ by translation by b. If we let h_∞ be the smallest positive integer such that $\gamma_\infty = \left(\begin{smallmatrix} 1 & h_\infty \\ 0 & 1 \end{smallmatrix}\right) \in \Gamma_\infty$ then

$$\Gamma_\infty = \langle \gamma_\infty, (-I)^{\varepsilon_\Gamma} \rangle.$$

By our previous discussion, there exists a matrix $\sigma_\mathfrak{a} \in \mathrm{SL}_2(\mathbb{Z})$ such that $\sigma_\mathfrak{a}\infty = \mathfrak{a}$. We call $\sigma_\mathfrak{a}$ a **scaling matrix** for $\mathfrak{a}$. Note that $\sigma_\mathfrak{a}$ is determined up to composition on the right by an element of Γ_∞. If $\mathfrak{a} = \infty$ then we may take $\sigma_\mathfrak{a} = I$ and in this case we suppress the dependence upon the scaling matrix. Scaling matrices are useful because they allow us to transfer information at the cusp $\mathfrak{a}$ to the cusp at ∞. We will use them to determine the stabilizer subgroups of other cusps. If $\sigma_\mathfrak{a}$ is a scaling matrix for the cusp $\mathfrak{a}$ then $\Gamma_\mathfrak{a} = \sigma_\mathfrak{a}\Gamma_\infty\sigma_\mathfrak{a}^{-1} \cap \Gamma$. Whence

$$\Gamma_\mathfrak{a} = \langle \gamma_\mathfrak{a}, (-I)^{\varepsilon_\Gamma} \rangle,$$

where $\gamma_{\mathfrak{a}}$ is of the form

$$\gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}} \begin{pmatrix} 1 & h_{\mathfrak{a}} \\ 0 & 1 \end{pmatrix} \sigma_{\mathfrak{a}}^{-1},$$

for some positive integer $h_{\mathfrak{a}}$. We call $h_{\mathfrak{a}}$ the **width** of the cusp $\mathfrak{a}$. Equivalently, $h_{\mathfrak{a}}$ is the shift of the minimal translation belonging to $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$. Also observe that $h_{\mathfrak{a}}$ does not depend on the choice of scaling matrix. We will set

$$\gamma_{\infty}^{(h_{\mathfrak{a}})} = \begin{pmatrix} 1 & h_{\mathfrak{a}} \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \Gamma_{\infty}^{(h_{\mathfrak{a}})} = \langle \gamma_{\infty}^{(h_{\mathfrak{a}})}, (-I)^{\varepsilon_\Gamma} \rangle,$$

so that

$$\Gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}} \Gamma_{\infty}^{(h_{\mathfrak{a}})} \sigma_{\mathfrak{a}}^{-1}.$$

In particular, $\gamma_{\infty}^{(h_\infty)} = \gamma_\infty$ and $\Gamma_{\infty}^{(h_\infty)} = \Gamma_\infty$.

Remark 1.12. Suppose $\mathfrak{a}'$ is a cusp that belongs to the same cusp class at $\mathfrak{a}$. Then $\mathfrak{a}' = \gamma\mathfrak{a}$ for some $\gamma \in \Gamma$. It follows that $\gamma\sigma_{\mathfrak{a}}$ is a scaling matrix for the cusp $\mathfrak{a}'$. So $\Gamma\sigma_{\mathfrak{a}}\Gamma_\infty$ only depends upon the cusp class of $\mathfrak{a}$ which is $\Gamma\sigma_{\mathfrak{a}}\infty$. Moreover, it is easy to see that $\Gamma_{\mathfrak{a}'} = \gamma\Gamma_{\mathfrak{a}}\gamma^{-1}$ and $\Gamma_{\mathfrak{a}'} = \gamma\sigma_{\mathfrak{a}}\Gamma_\infty\sigma_{\mathfrak{a}}^{-1}\gamma^{-1} \cap \Gamma$. Whence $\gamma_{\mathfrak{a}'} = \gamma\gamma_{\mathfrak{a}}\gamma^{-1}$ and therefore $h_{\mathfrak{a}} = h_{\mathfrak{a}'}$ because the width of a cusp does not depend upon the choice of scaling matrix. This means that the width of $\mathfrak{a}$ only depends upon its cusp class and so the same is true for $\gamma_{\infty}^{(h_{\mathfrak{a}})}$ and $\Gamma_{\infty}^{(h_{\mathfrak{a}})}$.

If Γ is of level N then N is the smallest positive integer such that $\Gamma(N) \subseteq \Gamma$. This means $\left(\begin{smallmatrix} 1 & N \\ 0 & 1 \end{smallmatrix}\right)$ is the minimal translation guaranteed to belong to Γ. Since principal congruence subgroups are normal in the modular group, we have $\Gamma(N) \subseteq \Gamma_{\infty}^{(h_{\mathfrak{a}})}$ and so the aforementioned translation must be a power of $\gamma_{\mathfrak{a}}$ which means $h_{\mathfrak{a}} \mid N$.

We say Γ is **reduced** at $\mathfrak{a}$ if $h_{\mathfrak{a}} = 1$. For example, the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, and $\Gamma(N)$ are all reduced at ∞. The width of the cusps are also related to the index d_Γ in a simple manner.

Proposition 1.13. *For any congruence subgroup Γ and any cusp $\mathfrak{a}$ of $\Gamma\backslash\mathbb{H}$, there are $h_{\mathfrak{a}}$ many cosets of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ whose elements are scaling matrices for cusps in the cusp class of $\mathfrak{a}$. In particular,*

$$\sum_{\mathfrak{a} \in \Gamma\backslash\hat{\mathbb{Q}}} h_{\mathfrak{a}} = d_\Gamma.$$

Proof. Let $\sigma_{\mathfrak{a}}$ be a scaling matrix for a cusp $\mathfrak{a}$. It follows from Remark 1.12 that the cusp classes of $\Gamma\backslash\mathbb{H}$ are in bijection with the double cosets of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})/\Gamma_\infty$ where a cusp class $\Gamma\sigma_{\mathfrak{a}}\infty$ corresponds to the double coset $\Gamma\sigma_{\mathfrak{a}}\Gamma_\infty$ under this bijection. As every element of $\mathrm{SL}_2(\mathbb{Z})$ is a scaling matrix for some cusp, consider the surjection

$$\phi : \Gamma\backslash\mathrm{SL}_2(\mathbb{Z}) \to \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})/\Gamma_\infty \qquad \Gamma\sigma_{\mathfrak{a}} \mapsto \Gamma\sigma_{\mathfrak{a}}\Gamma_\infty.$$

We will compute the number of cosets in a fiber of ϕ. Suppose $\mathfrak{a}$ and $\mathfrak{a}'$ are representatives of the same cusp class so that $\Gamma\sigma_{\mathfrak{a}}$ and $\Gamma\sigma_{\mathfrak{a}'}$ belong to the fiber of $\Gamma\sigma_{\mathfrak{a}}\Gamma_\infty = \Gamma\sigma_{\mathfrak{a}'}\Gamma_\infty$. Then $\sigma_{\mathfrak{a}'} = \gamma\sigma_{\mathfrak{a}}\gamma_\infty^n$ for some $\gamma \in \Gamma$ and integer n. Whence $\Gamma\sigma_{\mathfrak{a}} = \Gamma\sigma_{\mathfrak{a}'}$ if and only if $\Gamma\sigma_{\mathfrak{a}} = \Gamma\sigma_{\mathfrak{a}}\gamma_\infty^n$ which is to say that $\gamma_\infty^n \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$. As $\gamma_\infty^n \in \Gamma_\infty$, this latter condition is further equivalent to $\gamma_\infty^n \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{a} \cap \Gamma_\infty = \sigma_\mathfrak{a}^{-1}\Gamma_\mathfrak{a}\sigma_\mathfrak{a} = \Gamma_\infty^{(h_\mathfrak{a})}$ which happens if and only if $h_\mathfrak{a} \mid n$. So the cosets $\Gamma\sigma_\mathfrak{a}$ and $\Gamma\sigma_{\mathfrak{a}'}$ are equal if and only if $h_\mathfrak{a} \mid n$. This means there are $h_\mathfrak{a}$ cosets in the fiber of $\Gamma\sigma_\mathfrak{a}\Gamma_\infty$. In other words, there are $h_\mathfrak{a}$ many cosets of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ whose elements are scaling matrices for cusps in the cusp class of $\mathfrak{a}$. This proves the first statement. For the second, the sum of the number of cosets in the fibers of ϕ is equal to the order of $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ since ϕ is surjective. But this means

$$\sum_{\mathfrak{a}\in\Gamma\backslash\hat{\mathbb{Q}}} h_\mathfrak{a} = d_\Gamma,$$

proving the second statement. □

Let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma\backslash\mathbb{H}$ with scaling matrices $\sigma_\mathfrak{a}$ and $\sigma_\mathfrak{b}$ respectively. It will be essential to have a double coset decomposition for sets of the form $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$. This is known as the **Bruhat decomposition** for $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$.

Theorem (Bruhat decomposition). *Let Γ be a congruence subgroup and let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma\backslash\mathbb{H}$ with scaling matrices $\sigma_\mathfrak{a}$ and $\sigma_\mathfrak{b}$ respectively. Then we have the disjoint decomposition*

$$\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b} = \Omega_\infty \cup \left(\bigcup_{\substack{c\in\mathbb{Z}_+ \\ d \pmod{ch_\mathfrak{b}}}} \Omega_{c/d} \right),$$

where

$$\Omega_\infty = \Gamma_\infty^{(h_\mathfrak{a})} w_\infty \Gamma_\infty^{(h_\mathfrak{b})} \quad \text{and} \quad \Omega_{c/d} = \Gamma_\infty^{(h_\mathfrak{a})}\omega_{c/d}\Gamma_\infty^{(h_\mathfrak{b})},$$

with $\omega_\infty = \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right) \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ and $\omega_{c/d} = \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ if such matrices exists otherwise Ω_∞ and $\Omega_{c/d}$ are empty respectively. In the former case for Ω_∞, ω_∞ exists if and only if $\mathfrak{a}$ and $\mathfrak{b}$ are in the same cusp class in which case $h_\mathfrak{a} = h_\mathfrak{b}$. In the former case for $\Omega_{c/d}$, a and d are determined modulo $ch_\mathfrak{a}$ and $ch_\mathfrak{b}$ respectively and $\Omega_{c/d}$ does not depend on the entry a of $\omega_{c/d}$.

Proof. As $\Gamma_\infty^{(h_\mathfrak{a})} = \sigma_\mathfrak{a}^{-1}\Gamma_\mathfrak{a}\sigma_\mathfrak{a}$ and $\Gamma_\infty^{(h_\mathfrak{b})} = \sigma_\mathfrak{b}^{-1}\Gamma_\mathfrak{b}\sigma_\mathfrak{b}$, short computations show that $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ is invariant under multiplication from the left or right by elements of $\Gamma_\infty^{(h_\mathfrak{a})}$ and $\Gamma_\infty^{(h_\mathfrak{b})}$ respectively. This proves Ω_∞ and $\Omega_{c/d}$ are subsets of $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$. Now any $\omega_\infty \in \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ satisfies $\omega_\infty = \sigma_\mathfrak{a}^{-1}\gamma\sigma_\mathfrak{b}$ for some $\gamma \in \Gamma$. As $\omega_\infty\infty = \infty$, a short computation shows

$$\gamma\mathfrak{b} = \mathfrak{a}.$$

This means $\mathfrak{a}$ and $\mathfrak{b}$ belong to the same cusp class. Conversely, if $\mathfrak{a}$ and $\mathfrak{b}$ are in the same cusp class then $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ contains $\Gamma_\infty^{(h_\mathfrak{a})} = \Gamma_\infty^{(h_\mathfrak{b})}$ and we may take ω_∞ to be any such matrix. This proves that a ω_∞ exists if and only if $\mathfrak{a}$ and $\mathfrak{b}$ are in the same cusp class in which case $h_\mathfrak{a} = h_\mathfrak{b}$ by Remark 1.12. In this case, we may write

$$\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b} = \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{a}\omega_\infty \quad \text{and} \quad \sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b} = \omega_\infty\sigma_\mathfrak{b}^{-1}\Gamma\sigma_\mathfrak{b},$$

from which we see that the matrices in $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ acting by translation are exactly Ω_∞. Every other matrix in $\sigma_\mathfrak{a}^{-1}\Gamma\sigma_\mathfrak{b}$ belongs to one of the double cosets $\Omega_{c/d}$. The relation

$$\gamma_\infty^{(h_\mathfrak{a})}\omega_{c/d}\gamma_\infty^{(h_\mathfrak{b})} = \begin{pmatrix} a + ch_\mathfrak{a} & * \\ c & d + ch_\mathfrak{b} \end{pmatrix},$$

shows that $\Omega_{c/d}$ is generated uniquely by a matrix of the form $\omega_{c/d} = \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$ where a and d are determined modulo $ch_{\mathfrak{a}}$ and $ch_{\mathfrak{b}}$ respectively. To see that $\Omega_{c/d}$ does not depend on the entry a of $\omega_{c/d}$, let $\omega'_{c/d} = \left(\begin{smallmatrix} a' & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$. Write $\omega_{c/d} = \sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}}$ and $\omega'_{c/d} = \sigma_{\mathfrak{a}}^{-1}\gamma'\sigma_{\mathfrak{b}}$ for some $\gamma, \gamma' \in \Gamma$. The relations

$$\omega_{c/d}(\omega'_{c/d})^{-1} = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \quad \text{and} \quad \gamma(\gamma')^{-1} = \sigma_{\mathfrak{a}}\omega_{c/d}(\omega'_{c/d})^{-1}\sigma_{\mathfrak{a}}^{-1},$$

imply $\gamma(\gamma')^{-1} \in \Gamma_{\mathfrak{a}}$. So $\omega_{c/d}(\omega'_{c/d})^{-1} \in \Gamma_\infty^{(h_{\mathfrak{a}})}$ whence $\Gamma_\infty^{(h_{\mathfrak{a}})}\omega_{c/d}\Gamma_\infty^{(h_{\mathfrak{b}})} = \Gamma_\infty^{(h_{\mathfrak{a}})}\omega'_{c/d}\Gamma_\infty^{(h_{\mathfrak{b}})}$. □

As the double cosets Ω_∞ and $\Omega_{c/d}$ in the Bruhat decomposition may very well be empty, we would like to track exactly which are not. To do this, let $\mathcal{C}_{\mathfrak{a},\mathfrak{b}}$ and $\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$ be the sets

$$\mathcal{C}_{\mathfrak{a},\mathfrak{b}} = \left\{ c \in \mathbb{Z}_+ : \begin{pmatrix} * & * \\ c & * \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}} \right\} \quad \text{and} \quad \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) = \left\{ d \in \mathbb{Z}/ch_{\mathfrak{b}}\mathbb{Z} : \begin{pmatrix} * & * \\ c & d \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}} \right\},$$

where $\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$ is defined for $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$. These sets are constructed so that $\mathcal{C}_{\mathfrak{a},\mathfrak{b}}$ and $\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$ run over precisely the pairs (c, d) for which $\Omega_{c/d}$ is nonempty. Indeed, we may write

$$\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}} = \Omega_\infty \cup \left(\bigcup_{\substack{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}} \\ d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)}} \Omega_{c/d} \right),$$

where none of the double cosets $\Omega_{c/d}$ are empty.

1.4 Geometric Exponential Sums

We will now introduce two type of exponential sum associated to pairs of cusps. These are types of Kloosterman and Salié sums. We being with the Kloosterman sums. For any $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$ and integers m and n, the **Kloosterman sum** $K_{\mathfrak{a},\mathfrak{b}}(m, n, c)$ relative to $\mathfrak{a}$ and $\mathfrak{b}$ is defined by

$$K_{\mathfrak{a},\mathfrak{b}}(m, n, c) = \sum_{\substack{d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} e\left(\frac{ma}{ch_{\mathfrak{a}}} + \frac{nd}{ch_{\mathfrak{b}}} \right).$$

This sum is well-defined since a and d are determined modulo $ch_{\mathfrak{a}}$ and $ch_{\mathfrak{b}}$ respectively by the Bruhat decomposition. If $m = n = 0$ then

$$K_{\mathfrak{a},\mathfrak{b}}(0, 0, c) = |\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)|.$$

The Salié sums are defined in a similar manner. For any $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$, integers m and n, and Dirichlet character χ of conductor $q \mid c$, the **Salié sum** $S_{\chi,\mathfrak{a},\mathfrak{b}}(m, n, c)$ relative to $\mathfrak{a}$ and $\mathfrak{b}$ is defined by

$$S_{\chi,\mathfrak{a},\mathfrak{b}}(m, n, c) = \sum_{\substack{d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right) \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} \chi(d) e\left(\frac{ma}{ch_{\mathfrak{a}}} + \frac{nd}{ch_{\mathfrak{b}}} \right).$$

Similar as before, this sum is well-defined since a and d are determined modulo $ch_{\mathfrak{a}}$ and $ch_{\mathfrak{b}}$ respectively by the Bruhat decomposition and that the conductor of χ divides c. The usual Kloosterman and Salié sums can be obtained from those relative to cusps. Indeed, consider the case where $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ and $\mathfrak{a} = \mathfrak{b} = \infty$. The Bruhat decomposition for $\mathrm{SL}_2(\mathbb{Z})$ shows that the entires a and d in $\omega_{c/d} = \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right)$ are determined modulo c and so we may take $d = a^{-1}$ because $\det(\omega_{c/d}) = 1$. Also, $\mathcal{C}_{\infty,\infty} = \mathbb{Z}_+$ and $\mathcal{D}_{\infty,\infty}(c) = (\mathbb{Z}/c\mathbb{Z})^*$ because if $\gamma = \left(\begin{smallmatrix} * & * \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ we must have $(c,d) = 1$ as $\det(\gamma) = 1$ and every such pair gives rise to a matrix in $\mathrm{SL}_2(\mathbb{Z})$. Whence

$$K_{\infty,\infty}(n,m,c) = K(n,m,c) \quad \text{and} \quad S_{\chi,\infty,\infty}(n,m,c) = S_\chi(n,m,c),$$

which are the usual Kloosterman and Salié sums respectively. In these cases, we will suppress the dependencies upon the cusps.

1.5 The Hyperbolic Measure

We will need to integrate over various subsets of $\mathbb{H}$ and so we require a measure on this space. Our choice is the **hyperbolic measure** $d\mu$ given by

$$d\mu = \frac{dx\,dy}{y^2}.$$

This is the just the natural Lebesgue measure on $\mathbb{C}$ restricted to $\mathbb{H}$ and scaled by a factor of $\frac{1}{y^2}$. This allows for Lebesgue integration on $\mathbb{H}$ by defining

$$\int_{\mathbb{H}} f(z)\,d\mu,$$

for any complex function f on $\mathbb{H}$. The most important property about the hyperbolic measure is that it is $\mathrm{GL}_2^+(\mathbb{R})$-invariant.

Proposition 1.14. *The hyperbolic measure is* $\mathrm{GL}_2^+(\mathbb{R})$*-invariant.*

Proof. Let $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$. Since αz is a Möbius transformation it satisfies the Cauchy-Riemann equations. So for the change of variables $z \mapsto \alpha z$, the absolute value of the Jacobian determinant is the absolute value of the square of the derivative of αz. A short computation shows that this is $\frac{\det(\alpha)^2}{|cz+d|^4}$. From this fact and Equation (1), a short computation shows

$$\mu(\alpha z) = \mu(z).$$

This means $d\mu$ is $\mathrm{GL}_2^+(\mathbb{R})$-invariant. $\square$

Invariance of the hyperbolic measure will allow for integration over quotients. In particular, we will need to integrate over $\Gamma\backslash\mathbb{H}$ and $\Gamma_\infty\backslash\mathbb{H}$ which admit the fundamental domains

$$\mathcal{F}_\Gamma \quad \text{and} \quad \{z \in \mathbb{H} : 0 \le \mathrm{Re}(z) \le h_\infty \text{ and } \mathrm{Im}(z) > 0\},$$

respectively with the former following from Proposition 1.9 and the latter being clear since Γ_∞ is generated by the translation γ_∞ up to sign. Accordingly, we define

$$\int_{\Gamma\backslash\mathbb{H}} f(z)\,d\mu = \int_{\mathcal{F}_\Gamma} f(z)\,d\mu \quad \text{and} \quad \int_{\Gamma_\infty\backslash\mathbb{H}} f(z)\,d\mu = \int_0^\infty \int_0^{h_\infty} f(z)\,\frac{dx\,dy}{y^2},$$

for any Γ-invariant complex function f on $\mathbb{H}$. These integrals are well-defined because f and the hyperbolic measure are Γ-invariant. As a special case of the first integral, the **volume** V_Γ of $\Gamma\backslash\mathbb{H}$ is defined by

$$V_\Gamma = \int_{\Gamma\backslash\mathbb{H}} d\mu.$$

If $\mathcal{F}_\Gamma = \mathcal{F}$ then we write $V_\Gamma = V$. Using Proposition 1.7 to write this integral explicitly in the case of the modular group, a short computation shows

$$V = \int_{-\frac{1}{2}}^{\frac{1}{2}} \int_{\sqrt{1-x^2}}^\infty \frac{dy\,dx}{y^2} = \frac{\pi}{3}.$$

Therefore the volume of the standard fundamental domain with respect to the hyperbolic measure is finite despite this region being unbounded. There is also a simple relation between V_Γ and d_Γ given by

$$V_\Gamma = d_\Gamma V, \tag{3}$$

which follows immediately from Proposition 1.9. Therefore the volume of any modular curve is finite. A particularly nice application of this fact is that the former type of integrals over $\Gamma\backslash\mathbb{H}$ are absolutely convergent provided their integrands are bounded. As for another fact, the volume is invariant under conjugation. Indeed, let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ and suppose $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ so that it is a congruence subgroup by Lemma 1.3. A short computation using the change of variables $z \mapsto \alpha^{-1}z$ shows

$$V_{\alpha^{-1}\Gamma\alpha} = V_\Gamma. \tag{4}$$

It turns out that the integrals

$$\int_{\Gamma\backslash\mathbb{H}} f(z)\,d\mu \quad \text{and} \quad \int_{\Gamma_\infty\backslash\mathbb{H}} f(z)\,d\mu,$$

are related. This is known as the **unfolding/folding method**.

Theorem (Unfolding/folding method). *Let Γ be a congruence subgroup. If f is a complex function on $\mathbb{H}$ then*

$$\int_{\Gamma\backslash\mathbb{H}} \sum_{\gamma\in\Gamma_\infty\backslash\Gamma} f(\gamma z)\,d\mu = \int_{\Gamma_\infty\backslash\mathbb{H}} f(z)\,d\mu,$$

provided either side is absolutely convergent.

Proof. If the left-hand side converges absolutely then we may interchange the sum and integral. Upon making the change of variables $z \mapsto \gamma^{-1}z$, the invariance of $d\mu$ implies that the integral takes the form

$$\sum_{\gamma \in \Gamma_\infty \backslash \Gamma} \int_{\gamma \mathcal{F}_\Gamma} f(z)\, d\mu.$$

The result follows as $\mathbb{H} = \bigcup_{\gamma \in \Gamma} \gamma \mathcal{F}$. Moreover, we can argue in reverse provided the right-hand side converges absolutely. This completes the proof. $\square$

The process of transforming the left-hand side into the right-hand side is called **unfolding** while transforming the right-hand into the left-hand side is called **folding**.

2 Modular Forms

For any complex z let

$$e(z) = e^{2\pi i z}.$$

Also, we will distinguish the primitive third root of unity $\omega = e\left(\frac{1}{3}\right)$. All sums and products over a quotient group will be consider to be over a complete set of representatives and we identify cosets with corresponding representatives. If an integer a is taken modulo m, for any positive integer m, we write $\overline{a}$ for the inverse of a.

2.1 Modularity

For all all $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{Q})$ and $z \in \mathbb{H}$, define $j(\alpha, z)$ by

$$j(\alpha, z) = (cz + d).$$

With this notation, Equation (1) becomes

$$\mathrm{Im}(\alpha z) = \det(\alpha) \frac{\mathrm{Im}(z)}{|j(\alpha, z)|^2}.$$

There is a very useful property that $j(\alpha, z)$ satisfies. To state it, let $\alpha' = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{Q})$. Then

$$\alpha\alpha' = \begin{pmatrix} a'a + b'c & a'b + b'b \\ c'a + d'c & c'b + d'd \end{pmatrix},$$

and from this identity, a short computation shows

$$j(\alpha'\alpha, z) = j(\alpha', \alpha z) j(\alpha, z).$$

This identity is called the **cocycle condition** for $j(\alpha, z)$. For any positive integer k and $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, we define the **slash operator** $|_k\alpha : C(\mathbb{H}) \to C(\mathbb{H})$ to be the linear operator given by

$$(f|_k\alpha)(z) = \det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} f(\alpha z).$$

In the case of $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, the slash operator takes the simplified form

$$(f|_k\gamma)(z) = j(\gamma, z)^{-k} f(\gamma z).$$

In any case, the cocycle condition implies that the slash operator is multiplicative as short computations show

$$(f|_k\alpha'\alpha)(z) = ((f|_k\alpha')|_k\alpha)(z) \quad \text{and} \quad ((fg)|_{k+\ell}\alpha)(z) = (f|_k\alpha)(z)(g|_\ell\alpha)(z),$$

for $f, g \in C(\mathbb{H})$, positive integers k and ℓ, and any $\alpha, \alpha' \in \mathrm{GL}_2^+(\mathbb{Q})$. For fixed k, an operator on $C(\mathbb{H})$ is said to be **invariant** if it commutes with the slash operators $|_k\alpha$ for every $\alpha \in \mathrm{SL}_2(\mathbb{Q})$.

We can now introduce modular forms. Let Γ be a congruence subgroup of level N and $\nu : \Gamma \to \mathbb{C}^*$ be a finite order homomorphism. We say that a function $f : \mathbb{H} \to \mathbb{C}$ is **modular form** on $\Gamma\backslash\mathbb{H}$ of **weight** k, **level** N, and **nebentypus** ν if the following properties are satisfied:

(i) f is holomorphic on $\mathbb{H}$.

(ii) $(f|_k\gamma)(z) = \nu(\gamma)f(z)$ for all $\gamma \in \Gamma$.

(iii) $(f|_k\gamma)(z) = O(1)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.

Furthermore, we say f is a **cusp form** if the additional property is satisfied:

(iv) For all cusps $\mathfrak{a}$ and positive y, we have

$$\int_0^{h_\mathfrak{a}} (f|_k\sigma_\mathfrak{a})(x + iy)\, dx = 0.$$

Property (ii) is called the **modularity condition** and we say f is **modular**. The modularity condition can equivalently be expressed as

$$f(\gamma z) = \nu(\gamma)j(\gamma, z)^k f(z).$$

We write n_ν for the order of the nebentypus so that ν takes its values in the n_ν-th roots of unity. Moreover, we say that ν is **regular** at a cusp $\mathfrak{a}$ if $\nu(\gamma_\mathfrak{a}) = 1$. Equivalently, $\nu(\gamma) = 1$ for every $\gamma \in \Gamma_\mathfrak{a}$. Property (iii) is called the **growth condition** and we say f is **holomorphic at the cusps**. By modularity, it suffices to verify the growth condition for a set of scaling matrices associated to a complete set of Γ-inequivalent cusps. Therefore we only have to check this condition for finitely many matrices since the number of cusp classes is finite. Also, the growth condition is equivalent to $(f|_k\alpha)(z) = O(1)$ for all $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. To see this, we require a lemma.

Lemma 2.1. *Every $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ can be written as $\alpha = \gamma\eta$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and $\eta \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\eta = \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$.*

Proof. Let r be a positive integer such that $r\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{Z})$. As $r\alpha$ has nonzero determinant, we cannot have $a = c = 0$. Whence we may set $a' = \frac{a}{(a,c)}$ and $c' = \frac{c}{(a,c)}$ so that a' and c' are relatively prime integers. Then there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $\gamma^{-1} = \left(\begin{smallmatrix} * & * \\ -c' & a' \end{smallmatrix}\right)$. Moreover, a short computation shows

$$\gamma^{-1}r\alpha = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}.$$

The claim is complete upon setting $\eta = \gamma^{-1}\alpha$. $\square$

For the equivalence, use this lemma to write $\alpha = \gamma\eta$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and $\eta \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\eta = \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$. The cocycle condition gives

$$(f|_k\alpha)(z) = \det(\eta)^{\frac{k}{2}} j(\eta, z)(f|_k\gamma)(\eta z),$$

and as $j(\eta, z)$ is a constant, we have $(f|_k\alpha)(z) = O(1)$ proving the forward implication. The reverse implication is clear since $\mathrm{SL}_2(\mathbb{Z}) \subset \mathrm{GL}_2^+(\mathbb{Q})$.

Now let $\sigma_{\mathfrak{a}}$ be a scaling matrix for the $\mathfrak{a}$ cusp so that $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$ is a congruence subgroup of the same level as Γ. Let $\gamma \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$ and write $\gamma = \sigma_{\mathfrak{a}}^{-1}\gamma'\sigma_{\mathfrak{a}}$ for some $\gamma' \in \Gamma$. Using modularity, a short computation shows

$$((f|_k\sigma_{\mathfrak{a}})|_k\gamma)(z) = \nu_{\sigma_{\mathfrak{a}}}(\gamma)(f|_k\sigma_{\mathfrak{a}})(z),$$

where we have set

$$\nu_{\sigma_{\mathfrak{a}}}(\gamma) = \nu(\sigma_{\mathfrak{a}}\gamma\sigma_{\mathfrak{a}}^{-1}).$$

It follows that $f|_k\sigma_{\mathfrak{a}}$ is a modular form on $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}\backslash\mathbb{H}$ of the same weight and level as f but with nebentypus $\nu_{\sigma_{\mathfrak{a}}}$ and is a cusp form if f is.

Remark 2.2. If f is a weight k modular form with nebentypus ν on $\Gamma\backslash\mathbb{H}$ and $-I \in \Gamma$ then modularity implies

$$f(z) = \nu(-I)(-1)^k f(z).$$

Therefore $\nu(-I) = (-1)^k$. This means ν cannot be the identity if k is odd. In particular, there are no nonzero modular forms of odd weight with trivial nebentypus on $\mathrm{SL}_2(\mathbb{Z})$.

Modular forms also admit Fourier series. First observe that modularity implies

$$(f|_k\sigma_{\mathfrak{a}})(z + h_{\mathfrak{a}}) = \nu(\gamma_{\mathfrak{a}})(f|_k\sigma_{\mathfrak{a}})(z),$$

which is to say $f|_k\sigma_{\mathfrak{a}}$ is $h_{\mathfrak{a}}$-periodic up to the constant factor $\nu(\gamma_{\mathfrak{a}})$. This means we only need to verify the growth condition as $y \to \infty$. In any case, as $f|_k\sigma_{\mathfrak{a}}$ is also smooth, it admits an absolutely and uniformly convergent Fourier series. So we may write

$$(f|_k\sigma_{\mathfrak{a}})(z) = \sum_{n\geq 0} a_{\mathfrak{a}}(y, n)e\left(\frac{nx}{h_{\mathfrak{a}}}\right),$$

where the sum is only over nonnegative n because holomorphy at the cusps forces $f|_k\sigma_{\mathfrak{a}}$ to be bounded. We can also simplify the Fourier coefficients $a_{\mathfrak{a}}(y,n)$. Indeed, as $f|_k\sigma_{\mathfrak{a}}$ is holomorphic it satisfies the Cauchy-Riemann equations so that

$$\frac{1}{2}\left(\frac{\partial f|_k\sigma_{\mathfrak{a}}}{\partial x}+i\frac{\partial f|_k\sigma_{\mathfrak{a}}}{\partial y}\right)=0.$$

Substituting in the Fourier series and equating coefficients we obtain the ODE

$$\frac{2\pi n}{h_{\mathfrak{a}}}a_{\mathfrak{a}}(y,n)+a_{\mathfrak{a}}'(y,n)=0.$$

Solving this ODE by separation of variables, there exist coefficients $a_{\mathfrak{a}}(n)$ such that

$$a_{\mathfrak{a}}(y,n)=a_{\mathfrak{a}}(n)e\left(\frac{n(iy)}{h_{\mathfrak{a}}}\right).$$

The coefficients $a_{\mathfrak{a}}(n)$ are the only part of the Fourier series depending on the cusp $\mathfrak{a}$ and scaling matrix $\sigma_{\mathfrak{a}}$. Using these coefficients instead, f admits a **Fourier series** at the $\mathfrak{a}$ cusp given by

$$(f|_k\sigma_{\mathfrak{a}})(z)=\sum_{n\geq 0}a_{\mathfrak{a}}(n)e\left(\frac{nz}{h_{\mathfrak{a}}}\right),$$

with **Fourier coefficients** $a_{\mathfrak{a}}(n)$. In the case $\mathfrak{a}=\infty$ and $\sigma_{\mathfrak{a}}=I$, we will drop these dependences by writing $f|_k\sigma_{\mathfrak{a}}=f$ and $a_{\mathfrak{a}}(n)=a(n)$. From property (iv), f is a cusp form if and only if $a_{\mathfrak{a}}(0)=0$ for every cusp $\mathfrak{a}$. This does not depend upon the choice of scaling matrix used for $f|_k\sigma_{\mathfrak{a}}$ because

$$(f|_k\sigma_{\mathfrak{a}}\gamma_\infty)(z)=(f|_k\sigma_{\mathfrak{a}})(z+h_\infty), \tag{5}$$

and so the constant term in the Fourier series are identical. However, the nonconstant Fourier coefficients may very well change. Strictly speaking, if $\sigma_{\mathfrak{a}}'=\sigma_{\mathfrak{a}}\gamma_\infty^h$ for some positive integer h then

$$(f|_k\sigma_{\mathfrak{a}}')(z)=\sum_{n\geq 0}a_{\mathfrak{a}}(n)e\left(\frac{nh}{h_{\mathfrak{a}}}\right)e\left(\frac{nz}{h_{\mathfrak{a}}}\right),$$

so that the n-th Fourier coefficient is multiplied by the root of unity $e\left(\frac{h}{h_{\mathfrak{a}}}\right)$. Of course, if Γ is reduced at $\mathfrak{a}$ then this root of unity is 1 and the Fourier coefficients are genuinely independent of the choice of scaling matrix.

We can also derive a bound for the Fourier coefficients of f at the $\mathfrak{a}$ cusp provided f is a cusp form. This bound is quite straightforward to deduce and turns out to almost be optimal. To this end, consider the function

$$\left|(f|_k\sigma_{\mathfrak{a}})(z)\mathrm{Im}(z)^{\frac{k}{2}}\right|,$$

which is $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$-invariant. It is also bounded on $\mathcal{F}_{\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}}$ because f is a cusp form and the $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$-invariance forces boundedness on $\mathbb{H}$. Now consider the integral

$$\int_0^{h_{\mathfrak{a}}}(f|_k\sigma_{\mathfrak{a}})(z)\mathrm{Im}(z)^{\frac{k}{2}}e\left(-\frac{nz}{h_{\mathfrak{a}}}\right)\,dx.$$

On the one hand, substitute the Fourier series into this integral and use absolute convergence to interchange the sum and the integral. In view of the identity

$$\int_0^h e^{\frac{2\pi i(n-m)x}{h}}\,dx = h\delta_{n-m,0}, \tag{6}$$

for any positive integer h and integers n and m, every term integrates to zero except the n-th term which is $a_{\mathfrak{a}}(n)h_{\mathfrak{a}}\mathrm{Im}(z)^{\frac{k}{2}}$. On the other hand, the integral is $O\left(e^{-\frac{2\pi ny}{h_{\mathfrak{a}}}}\right)$ via our previous remarks. This shows

$$a_{\mathfrak{a}}(n)\mathrm{Im}(z)^{\frac{k}{2}} \ll e^{-\frac{2\pi ny}{h_{\mathfrak{a}}}}.$$

Upon setting $y = \frac{1}{n}$, the right-hand side is uniformly bounded and we obtain

$$a_{\mathfrak{a}}(n) \ll n^{\frac{k}{2}}.$$

This estimate is known as the **Hecke bound** for modular forms. It follows from the Hecke bound that

$$(f|_k\sigma_{\mathfrak{a}})(z) = O\left(e^{-\frac{2\pi y}{h_{\mathfrak{a}}}}\right),$$

upon summing the resulting geometric series. This implies $f|_k\sigma_{\mathfrak{a}}$ exhibits exponential decay and we say that f exhibits **exponential decay at the cusps**. This is stronger than simply knowing $f|_k\sigma_{\mathfrak{a}}$ is bounded on $\mathbb{H}$ and tends to zero as $y \to \infty$ which is provided by the fact that f is a cusp form.

Many of the modular forms we will study will be over the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, or $\Gamma(N)$. To collect all of these cases, let Γ denote any of these congruence subgroups. There are a few important simplifications when working over Γ. The first, which we have already noted, is that Γ is reduced ∞. Therefore all of our modular forms will be 1-periodic. Second is that the nebentypus can be induced by a Dirichlet character χ modulo N. Let q be the conductor of χ so that $q \mid N$. For any $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma$ we know that $c \equiv 0 \pmod{N}$ whence a is the inverse of d modulo N and thus $(d,q) = 1$. We can define a function $\nu_\chi : \Gamma \to \mathbb{C}^*$ by

$$\nu_\chi(\gamma) = \chi(d).$$

If $\gamma' = \left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right) \in \Gamma$, then the relations

$$\gamma\gamma' \equiv \begin{pmatrix} * & * \\ 0 & d'd \end{pmatrix} \pmod{N} \quad \text{and} \quad \gamma^{-1} = \begin{pmatrix} d & * \\ 0 & a \end{pmatrix} \pmod{N},$$

show

$$\nu_\chi(\gamma\gamma') = \nu_\chi(\gamma)\nu_\chi(\gamma') \quad \text{and} \quad \nu_\chi(\gamma^{-1}) = \nu_\chi(\gamma)^{-1}.$$

respectively. In other words, ν_χ is a homomorphism and it is necessarily of finite order because χ is. Therefore ν_χ satisfies the assumptions of a nebentypus. It is also regular at ∞. In the case f has nebentypus ν_χ, we simply say that f has **character** χ and write

$$\chi(\gamma) = \nu_\chi(\gamma).$$

2.2 Poincaré and Eisenstein Series

Fix a positive integer N and let Γ denote any of the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, or $\Gamma(N)$. We will introduce two important classes of modular forms on $\Gamma\backslash\mathbb{H}$. These are the Poincaré series and Eisenstein series where the latter is a special case of the former. Let m be a nonnegative integer, $k \geq 3$, $\nu : \Gamma \to \mathbb{C}^*$ be a finite order homomorphism, and $\mathfrak{a}$ be a cusp of $\Gamma\backslash\mathbb{H}$ with scaling matrix $\sigma_{\mathfrak{a}}$. Also assume ν is regular at $\mathfrak{a}$. We define the m-th (modular) **Poincaré series** $P_{m,k,\nu,\mathfrak{a}}(z)$ of weight k with nebentypus ν on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{a}$ cusp by

$$P_{m,k,\nu,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}}\backslash\Gamma} \overline{\nu}(\gamma) \left(e\left(\frac{mz}{h_{\mathfrak{a}}}\right) \bigg|_k \sigma_{\mathfrak{a}}^{-1}\gamma \right)(z).$$

We call m the **index** of $P_{m,k,\nu,\mathfrak{a}}(z)$. If ν is trivial or $\mathfrak{a} = \infty$, we will drop these dependencies accordingly.

Remark 2.3. The reason why we restrict to $k \geq 3$ is because for $k = 0, 1, 2$ the Poincaré series need not converge for any $z \in \mathbb{H}$.

The Poincaré series $P_{m,k,\nu,\mathfrak{a}}(z)$ is well-defined as it is independent of the choice of representatives γ and $\sigma_{\mathfrak{a}}$. For the nebentypus, this is easily checked by direct computation using the fact that ν is regular at $\mathfrak{a}$. For the exponential factor, the set of representatives for $\sigma_{\mathfrak{a}}^{-1}\gamma$ is $\Gamma_\infty^{(h_{\mathfrak{a}})}\sigma_{\mathfrak{a}}^{-1}\gamma$ in view of $\Gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}}\Gamma_\infty^{(h_{\mathfrak{a}})}\sigma_{\mathfrak{a}}^{-1}$. So by multiplicativity of the slash operator, it suffices to show

$$\left(e\left(\frac{mz}{h_{\mathfrak{a}}}\right) \bigg|_k \gamma_\infty^{(h_{\mathfrak{a}})} \right)(z) = e\left(\frac{mz}{h_{\mathfrak{a}}}\right).$$

This is easily verified by direct computation. Whence $P_{m,k,\nu,\mathfrak{a}}(z)$ is well-defined. To see that $P_{m,k,\nu,\mathfrak{a}}(z)$ is holomorphic on $\mathbb{H}$, the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma$ and bounding term by term gives

$$P_{m,k,\nu,\mathfrak{a}}(z) \ll \sum_{(c,d) \in \mathbb{Z}_{0+}^2 - \{\mathbf{0}\}} \frac{1}{|cz+d|^k}.$$

This latter series is locally absolutely uniformly convergent for $z \in \mathbb{H}$ as $k \geq 3$. Hence $P_{m,k,\nu,\mathfrak{a}}(z)$ is holomorphic on $\mathbb{H}$. As for modularity, let $\eta \in \Gamma$. A short computation using the change of variables $\eta \mapsto \eta\gamma^{-1}$ shows

$$(P_{m,k,\nu,\mathfrak{a}}|_k\eta)(z) = \nu(\eta)P_{m,k,\nu,\mathfrak{a}}(z).$$

To verify the growth condition, we will need a technical lemma.

Lemma 2.4. *Let $a, b > 0$ and consider set*

$$S_{a,b} = \{z \in \mathbb{H} : |\mathrm{Re}(z)| \leq a \text{ and } \mathrm{Im}(z) \geq b\}.$$

Then there exists a $\delta \in (0,1)$ such that

$$|nz + m| \geq \delta|ni + m|,$$

for all pairs of integers (n, m) and all $z \in S_{a,b}$.

Proof. If $n = 0$ then the claim is obvious as any δ is sufficient. If $n \neq 0$ then it is equivalent to prove

$$\left| \frac{z + \frac{m}{n}}{i + \frac{m}{n}} \right| \geq \delta.$$

To this end, consider the function

$$f(z, r) = \left| \frac{z + r}{i + r} \right|,$$

for $z \in S_{a,b}$ and real r. We will prove the stronger statement $f(z, r) \geq \delta$. Clearly $f(z, r)$ is continuous on $S_{a,b} \times \mathbb{R}$ and it is also positive there since $z - r \neq 0$. A short computation shows

$$f(z, r)^2 \geq \frac{|z|^2 + r^2}{1 + r^2}.$$

Since $\frac{r^2}{1+r^2} \to 1$ as $r \to \pm\infty$, choose $Y > b$ such that $\frac{r^2}{1+r^2} \geq \frac{1}{4}$ for $|r| > Y$. Now let

$$S_{a,b}^Y = \{z \in \mathbb{H} : |\mathrm{Re}(z)| \leq a \text{ and } b \leq \mathrm{Im}(z) \leq Y\},$$

so that

$$f(z, r) \geq \frac{1}{2},$$

outside of $S_{a,b}^Y \times [-Y, Y]$. But this latter region is compact and so $f(z, r)$ attains a minimum $\delta' > 0$ on it. Setting $\delta = \min\left(\frac{1}{2}, \delta'\right)$ completes the proof. $\square$

We can now verify the growth condition for $P_{m,k,\nu,\mathfrak{a}}(z)$. Choose a cusp $\mathfrak{b}$ and an associated scaling matrix $\sigma_{\mathfrak{b}}$. Using the cocycle condition, Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$, and bounding term by term shows give

$$(P_{m,k,\nu,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) \ll \sum_{(c,d) \in \mathbb{Z}_{0+}^2 - \{\mathbf{0}\}} \frac{1}{|cz + d|^k}.$$

Now decompose this last sum as

$$\sum_{(c,d) \in \mathbb{Z}_{0+}^2 - \{\mathbf{0}\}} \frac{1}{|cz + d|^k} = \sum_{d \geq 1} \frac{1}{d^k} + \sum_{c \geq 1} \sum_{d \geq 0} \frac{1}{|cz + d|^k}.$$

Since the first sum on the right-hand side is bounded, it suffices to show that the double sum is too as $z \to \infty$. So let $\mathrm{Im}(z) \geq 1$ and δ be as in Lemma 2.4. Then for any positive integer N, we may write

$$\sum_{c \geq 1} \sum_{d \geq 0} \frac{1}{|cz + d|^k} \leq \sum_{c + d \leq N} \frac{1}{|cz + d|^k} + \frac{1}{\delta^k} \sum_{c + d > N} \frac{1}{|ci + d|^k}.$$

As $k \geq 3$, the second sum on the right-hand side is the tail of an absolute convergent series. Whence it tends to zero as $N \to \infty$. As for the first sum, it is finite and each term is bounded. Together this shows that the double sum is bounded as $z \to \infty$ verifying the growth condition. We collect this work in the following theorem:

Theorem 2.5. *The Poincaré series*

$$P_{m,k,\nu,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \overline{\nu}(\gamma) \left(e\left(\frac{mz}{h_{\mathfrak{a}}}\right) \Big|_k \sigma_{\mathfrak{a}}^{-1}\gamma \right)(z),$$

is a weight k modular forms with nebentypus ν on $\Gamma\backslash\mathbb{H}$.

In the case of index zero Poincaré series, we write $E_{k,\nu,\mathfrak{a}}(z) = P_{0,k,\nu,\mathfrak{a}}(z)$ and call $E_{k,\nu,\mathfrak{a}}(z)$ the (modular) **Eisenstein series** of weight k and nebentypus ν on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{a}$ cusp. It is defined by

$$E_{k,\nu,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \overline{\nu}(\gamma)(1|_k \sigma_{\mathfrak{a}}^{-1}\gamma)(z).$$

Just as for the Poincaré series, if ν is trivial or $\mathfrak{a} = \infty$, we will drop these dependencies accordingly. In fact, we obtain the following theorem as a special case of the former one:

Theorem 2.6. *The Eisenstein series*

$$E_{k,\nu,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \overline{\nu}(\gamma)(1|_k \sigma_{\mathfrak{a}}^{-1}\gamma)(z),$$

is a weight k modular forms with nebentypus ν on $\Gamma\backslash\mathbb{H}$.

We will now take $\nu = \chi$ for χ a Dirichlet character modulo N. Note that χ is always regular at ∞. In this simplified case, we can explicitly compute the Fourier series of the Poincaré series and Eisenstein series whose Fourier coefficients involve Salié sums, or Kloosterman sums when χ is trivial, relative to cusps. This will essentially follow from a careful application of Poisson summation. Unlike verifying that these series are modular forms, the arguments are slightly different to compute their Fourier series. We first deduce the Fourier series of the Poincaré series.

Proposition 2.7. *Let $\mathfrak{b}$ be a cusp of $\Gamma\backslash\mathbb{H}$ with scaling matrix $\sigma_{\mathfrak{b}}$. The Fourier series of $P_{m,k,\chi,\mathfrak{a}}(z)$ on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{b}$ cusp is given by*

$$\begin{aligned}(P_{m,k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) &= \delta_{\Gamma_{\mathfrak{a}},\Gamma_{\mathfrak{b}}} e\left(\frac{mz}{h_{\mathfrak{b}}}\right) \\ &\quad + \sum_{n \ge 1} \left(2\pi i^{-k} \left(\frac{\sqrt{nh_{\mathfrak{a}}}}{\sqrt{mh_{\mathfrak{b}}}}\right)^{k-1} \sum_{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}} \frac{S_{\chi,\mathfrak{a},\mathfrak{b}}(m,n,c)}{ch_{\mathfrak{b}}} J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c\sqrt{h_{\mathfrak{a}}h_{\mathfrak{b}}}}\right) \right) e\left(\frac{nz}{h_{\mathfrak{b}}}\right),\end{aligned}$$

provided the index m is positive.

Proof. By multiplicativity of the slash operator, write

$$(P_{m,k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \overline{\chi}(\gamma) j(\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}}, z)^{-k} e\left(\frac{m\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}} z}{h_{\mathfrak{a}}}\right).$$

From the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$, any $\gamma \in \Gamma_{\mathfrak{a}}\backslash\Gamma$ is of the form $\gamma = \left(\begin{smallmatrix} a & b \\ c & c\ell h_{\mathfrak{b}}+d \end{smallmatrix}\right)$ where $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$, $d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$, and $\ell \in \mathbb{Z}$. Furthermore, a short computation shows

$$\frac{az+b}{cz+(ch_{\mathfrak{b}}\ell+d)} = \frac{a}{c} - \frac{1}{c^2z+c^2h_{\mathfrak{b}}\ell+cd}.$$

Using these facts together with the periodicity of χ, we have

$$(P_{m,k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) = \delta_{\Gamma_{\mathfrak{a}},\Gamma_{\mathfrak{b}}} e\left(\frac{mz}{h_{\mathfrak{b}}}\right) + \sum_{\substack{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}} \\ d\in\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right)\in\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} \overline{\chi}(d) \sum_{\ell\in\mathbb{Z}} \frac{e\left(\frac{m}{h_{\mathfrak{a}}}\left(\frac{a}{c} - \frac{1}{c^2z+c^2h_{\mathfrak{b}}\ell+cd}\right)\right)}{(cz+ch_{\mathfrak{b}}\ell+d)^k}.$$

Now let

$$I_{c,d}(z) = \sum_{\ell\in\mathbb{Z}} \frac{e\left(\frac{m}{h_{\mathfrak{a}}}\left(\frac{a}{c} - \frac{1}{c^2z+c^2h_{\mathfrak{b}}\ell+cd}\right)\right)}{(cz+ch_{\mathfrak{b}}\ell+d)^k}.$$

We will apply the Poisson summation formula to $I_{c,d}(z)$. By the identity theorem it suffices to verify the claim for $z = iy$ with y positive. Accordingly, let $f(x)$ be given by

$$f(x) = \frac{e\left(\frac{m}{h_{\mathfrak{a}}}\left(\frac{a}{c} - \frac{1}{c^2(iy)+c^2h_{\mathfrak{b}}x+cd}\right)\right)}{(c(iy)+ch_{\mathfrak{b}}x+d)^k}.$$

Then $f(x)$ is Schwartz class. We will compute its Fourier transform

$$(\mathcal{F}f)(n) = \int_{-\infty}^{\infty} \frac{e\left(\frac{m}{h_{\mathfrak{a}}}\left(\frac{a}{c} - \frac{1}{c^2(iy)+c^2h_{\mathfrak{b}}x+cd}\right)\right)}{(c(iy)+ch_{\mathfrak{b}}x+d)^k} e(-nx)\,dx.$$

Complexify the integral to get

$$\int_{\mathrm{Im}(w)=0} \frac{e\left(\frac{m}{h_{\mathfrak{a}}}\left(\frac{a}{c} - \frac{1}{c^2(iy)+c^2h_{\mathfrak{b}}w+cd}\right)\right)}{(c(iy)+ch_{\mathfrak{b}}w+d)^k} e(-nw)\,dw,$$

and make the change of variables $w \mapsto w - \frac{d}{ch_{\mathfrak{b}}} - \frac{(iy)}{h_{\mathfrak{b}}}$ to obtain

$$e\left(\frac{ma}{ch_{\mathfrak{a}}} + \frac{nd}{ch_{\mathfrak{b}}} + \frac{n(iy)}{h_{\mathfrak{b}}}\right) \int_{\mathrm{Im}(w)=\frac{y}{h_{\mathfrak{b}}}} \frac{e\left(-\frac{m}{c^2h_{\mathfrak{a}}h_{\mathfrak{b}}w}\right)}{(ch_{\mathfrak{b}}w)^k} e(-nw)\,dw.$$

The integrand is meromorphic with a pole at $w = 0$ and has polynomial decay of order k provided n is nonpositive. So for these n, we conclude that the integral vanishes upon shifting the line of integration to $\mathrm{Im}(w) = T$ for positive T and taking the limit as $T \to \infty$. It remains to compute the integral for positive n. To do this, make the change of variables $w \mapsto -\frac{w}{2\pi in}$ to the last integral to rewrite it as

$$\frac{(-1)^k(2\pi in)^{k-1}}{(ch_{\mathfrak{b}})^k} \int_{\left(\frac{2\pi ty}{h_{\mathfrak{b}}}\right)} \frac{e^{w-\frac{4\pi^2mn}{c^2h_{\mathfrak{a}}h_{\mathfrak{b}}w}}}{w^k}\,dw.$$

Homotoping the line of integration to the Hankel contour about the negative real axis, we recognize the integral as the Schläflin integral representation for the J-Bessel function since m is positive. This gives

$$2\pi i^{-k}\left(\frac{\sqrt{nh_{\mathfrak{a}}}}{\sqrt{mh_{\mathfrak{b}}}}\right)^{k-1}\frac{1}{ch_{\mathfrak{b}}}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c\sqrt{h_{\mathfrak{a}}h_{\mathfrak{b}}}}\right).$$

So in total we obtain

$$(\mathcal{F}f)(n)=\begin{cases}\left(2\pi i^{-k}\left(\frac{\sqrt{nh_{\mathfrak{a}}}}{\sqrt{mh_{\mathfrak{b}}}}\right)^{k-1}\frac{e\left(\frac{ma}{ch_{\mathfrak{a}}}+\frac{nd}{ch_{\mathfrak{b}}}\right)}{ch_{\mathfrak{b}}}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c\sqrt{h_{\mathfrak{a}}h_{\mathfrak{b}}}}\right)\right)e\left(\frac{n(iy)}{h_{\mathfrak{b}}}\right) & \text{if } n\geq 1,\\ 0 & \text{if } n\leq 0.\end{cases}$$

Then by Poisson summation, we have

$$I_{c,r}(z)=\sum_{n\geq 1}\left(2\pi i^{-k}\left(\frac{\sqrt{nh_{\mathfrak{a}}}}{\sqrt{mh_{\mathfrak{b}}}}\right)^{k-1}\frac{e\left(\frac{ma}{ch_{\mathfrak{a}}}+\frac{nd}{ch_{\mathfrak{b}}}\right)}{ch_{\mathfrak{b}}}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c\sqrt{h_{\mathfrak{a}}h_{\mathfrak{b}}}}\right)\right)e\left(\frac{nz}{h_{\mathfrak{b}}}\right).$$

Plugging this back into the Poincaré series, interchanging the sums over n and c by absolute convergence, and collecting the exponential sum over a and d, we at last obtain

$$\begin{aligned}(P_{m,k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)&=\delta_{\Gamma\mathfrak{a},\Gamma\mathfrak{b}}e\left(\frac{mz}{h_{\mathfrak{b}}}\right)\\&\quad+\sum_{n\geq 1}\left(2\pi i^{-k}\left(\frac{\sqrt{nh_{\mathfrak{a}}}}{\sqrt{mh_{\mathfrak{b}}}}\right)^{k-1}\sum_{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}}\frac{S_{\chi,\mathfrak{a},\mathfrak{b}}(m,n,c)}{ch_{\mathfrak{b}}}J_{k-1}\left(\frac{4\pi\sqrt{mn}}{c\sqrt{h_{\mathfrak{a}}h_{\mathfrak{b}}}}\right)\right)e\left(\frac{nz}{h_{\mathfrak{b}}}\right).\end{aligned}$$

which is our desired Fourier series. □

An immediate consequence of this result, upon inspection of the Fourier series, is that the Poincaré series $(P_{m,k,\chi,\mathfrak{a}}|\sigma_{\mathfrak{b}})(z)$ of positive index are cusp forms.

In a similar manner to the Poincaré series, we can obtain the Fourier series of the Eisenstein series too. The proof is only slightly different in that we will pass by the pole of the integrand instead of deforming to the Hankel contour about the negative real axis.

Proposition 2.8. *Let $\mathfrak{b}$ be a cusp of $\Gamma\backslash\mathbb{H}$ with scaling matrix $\sigma_{\mathfrak{b}}$. The Fourier series of $E_{k,\chi,\mathfrak{a}}(z)$ on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{b}$ cusp is given by*

$$(E_{k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)=\delta_{\Gamma\mathfrak{a},\Gamma\mathfrak{b}}+\sum_{n\geq 1}\left(\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)}\sum_{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}}\frac{S_{\chi}(0,n,c)}{(ch_{\mathfrak{b}})^k}\right)e\left(\frac{nz}{h_{\mathfrak{b}}}\right).$$

Proof. By multiplicativity of the slash operator, write

$$(E_{k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)=\sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma}\overline{\chi}(\gamma)j(\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{b}},z)^{-k}.$$

From the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$, any $\gamma \in \Gamma_{\mathfrak{a}}\backslash\Gamma$ is of the form $\gamma = \left(\begin{smallmatrix} a & b \\ c & c\ell h_{\mathfrak{b}}+d \end{smallmatrix}\right)$ where $c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}$, $d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c)$, and $\ell \in \mathbb{Z}$. Furthermore, a short computation shows

$$\frac{az+b}{cz+(ch_{\mathfrak{b}}\ell+d)} = \frac{a}{c} - \frac{1}{c^2z+c^2h_{\mathfrak{b}}\ell+cd}.$$

Using these facts together with the periodicity of χ, we have

$$(E_{k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) = \delta_{\Gamma_{\mathfrak{a}},\Gamma_{\mathfrak{b}}} + \sum_{\substack{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}} \\ d\in\mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \left(\begin{smallmatrix} a & * \\ c & d \end{smallmatrix}\right)\in\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}}} \overline{\chi}(d)\sum_{\ell\in\mathbb{Z}} \frac{1}{(cz+ch_{\mathfrak{b}}\ell+d)^k}.$$

Now let

$$I_{c,d}(z) = \sum_{\ell\in\mathbb{Z}} \frac{1}{(cz+ch_{\mathfrak{b}}\ell+d)^k}.$$

We will apply the Poisson summation formula to $I_{c,d}(z)$. By the identity theorem it suffices to verify the claim for $z = iy$ with y positive. Accordingly, let $f(x)$ be given by

$$f(x) = \frac{1}{(c(iy)+ch_{\mathfrak{b}}x+d)^k}.$$

Then $f(x)$ is Schwartz class. We will compute its Fourier transform

$$(\mathcal{F}f)(n) = \int_{-\infty}^{\infty} \frac{1}{(c(iy)+ch_{\mathfrak{b}}x+d)^k} e(-nx)\,dx.$$

Complexify the integral to get

$$\int_{\mathrm{Im}(w)=0} \frac{1}{(c(iy)+ch_{\mathfrak{b}}w+d)^k} e(-nw)\,dw,$$

and make the change of variables $w \mapsto w - \frac{d}{ch_{\mathfrak{b}}} - \frac{(iy)}{h_{\mathfrak{b}}}$ to obtain

$$e\left(\frac{nd}{ch_{\mathfrak{b}}} + \frac{n(iy)}{h_{\mathfrak{b}}}\right)\int_{\mathrm{Im}(w)=\frac{y}{h_{\mathfrak{b}}}} \frac{1}{(ch_{\mathfrak{b}}w)^k} e(-nw)\,dw.$$

The integrand is meromorphic with a pole at $w = 0$ and has polynomial decay of order k provided n is nonpositive. So for these n, we conclude that the integral vanishes upon shifting the line of integration to $\mathrm{Im}(w) = T$ for positive T and taking the limit as $T \to \infty$. It remains to compute the integral for positive n. To do this, make the change of variables $w \mapsto -\frac{w}{2\pi in}$ to the last integral to rewrite it as

$$\frac{(-1)^k(2\pi in)^{k-1}}{(ch_{\mathfrak{b}})^k}\int_{\left(\frac{2\pi ty}{h_{\mathfrak{b}}}\right)} \frac{e^w}{w^k}\,dw.$$

The integrand of the remaining integral is meromorphic with a pole of order k at $w = 0$. From the Laurent series of $\frac{e^w}{w^k}$, the residue is easily seen to be $\Gamma(k)$. So upon shifting the line of integration to $(-\varepsilon)$, we pass by this pole and obtain

$$\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)(ch_{\mathfrak{b}})^k} + \frac{(-1)^k(2\pi i n)^{k-1}}{(ch_{\mathfrak{b}})^k}\int_{(-\varepsilon)} \frac{e^w}{w^k}\,dw.$$

This last integrand has polynomial decay of order k. Whence we may again conclude that the integral vanishes upon shifting the line of integration to $\mathrm{Im}(w) = -T$ for positive T and taking the limit as $T \to \infty$. So in total we obtain

$$(\mathcal{F}f)(n) = \begin{cases} \left(\frac{(-2\pi i)^k t^{k-1}}{\Gamma(k)} \frac{e\left(\frac{nd}{ch_{\mathfrak{b}}}\right)}{(ch_{\mathfrak{b}})^k}\right) e\left(\frac{n(iy)}{h_{\mathfrak{b}}}\right) & \text{if } n \geq 1, \\ 0 & \text{if } n \leq 0. \end{cases}$$

Then by Poisson summation, we have

$$I_{c,r}(z) = \sum_{n\geq 1}\left(\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)} \frac{e\left(\frac{nd}{ch_{\mathfrak{b}}}\right)}{(ch_{\mathfrak{b}})^k}\right) e\left(\frac{nz}{h_{\mathfrak{b}}}\right).$$

Plugging this back into the Poincaré series, interchanging the sums over n and c by absolute convergence, and collecting the exponential sum over d, we at last obtain

$$(E_{k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) = \delta_{\Gamma\mathfrak{a},\Gamma\mathfrak{b}} + \sum_{n\geq 1}\left(\frac{(-2\pi i)^k n^{k-1}}{\Gamma(k)} \sum_{c\in\mathcal{C}_{\mathfrak{a},\mathfrak{b}}} \frac{S_\chi(0,n,c)}{(ch_{\mathfrak{b}})^k}\right) e\left(\frac{nz}{h_{\mathfrak{b}}}\right).$$

which is our desired Fourier series. □

An interesting observation, upon direct inspection of the Fourier series, is that the Eisenstein series $(E_{k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z)$ are cusp forms unless $\mathfrak{a}$ is in the same cusp class as $\mathfrak{b}$. Also, as Γ was one of the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, or $\Gamma(N)$, these Fourier series computations show that there exist cusp and non-cusp modular forms on these congruence subgroups. In particular, there exist modular forms on the Hecke congruence subgroups for every level.

2.3 Inner Product Spaces of Modular Forms

Fix a positive integer N and let Γ denote any of the congruence subgroups $\Gamma_0(N)$, $\Gamma_1(N)$, or $\Gamma(N)$. Let $\mathcal{M}_k(\Gamma,\nu)$ denote the complex vector space of all weight k modular forms with nebentypus ν on $\Gamma\backslash\mathbb{H}$, write $\mathcal{S}_k(\Gamma,\nu)$ for the subspace of cusp forms, and write $\mathcal{E}_k(\Gamma,\nu)$ for the subspace spanned by the Eisenstein series. If ν is trivial we will suppress this dependence. Note that if Γ_1 and Γ_2 are two congruence subgroups such that $\Gamma_1 \subseteq \Gamma_2$ then we have the inclusions

$$\mathcal{M}_k(\Gamma_2,\nu) \subseteq \mathcal{M}_k(\Gamma_1,\nu), \quad \mathcal{S}_k(\Gamma_2,\nu) \subseteq \mathcal{S}_k(\Gamma_1,\nu), \quad \text{and} \quad \mathcal{E}_k(\Gamma_2,\nu) \subseteq \mathcal{E}_k(\Gamma_1,\nu).$$

The vector space $\mathcal{M}_k(\Gamma,\nu)$ turns out to be finite dimensional. This will be accomplished by showing that any $f \in \mathcal{M}_k(\Gamma,\nu)$ is determined by finitely many of its Fourier coefficients. In fact, this will follow from an exact formula relating the weight k, the index d_Γ, and the zeros of f. This is known as the valence formula and we require some notation to state it. For any $z \in \mathbb{H}$, let $v_z(f)$ be the index of the first nonzero term in the Taylor series of f at z. Equivalently, $v_z(f)$ is the order of vanishing of f at z. Note that $v_z(f)$ is necessarily nonnegative as f is holomorphic and

$$v_{\gamma z}(f) = v_z(f),$$

for any $\gamma \in \Gamma$ by modularity. For a cusp $\mathfrak{a}$ of $\Gamma\backslash\mathbb{H}$, we let $v_{\mathfrak{a}}(f)$ be the index of the first nonzero term in the Fourier series of f at the $\mathfrak{a}$ cusp. That is, $v_{\mathfrak{a}}(f)$ is the order of vanishing of $f|_k\sigma_{\mathfrak{a}}$ at ∞. To show $v_{\mathfrak{a}}(f)$ is well-defined, we need to verify that it is independent of the choice of scaling matrix $\sigma_{\mathfrak{a}}$. This follows from Equation (5) since scaling matrices are determined up to composition on the right by an element of Γ_∞. Also, $v_{\mathfrak{a}}(f)$ is again nonnegative by the holomorphy of f and

$$v_{\mathfrak{a}'}(f) = v_{\mathfrak{a}}(f),$$

for any cusp $\mathfrak{a}'$ in the same cusp class as $\mathfrak{a}$ by modularity.

Our method for proving the valance formula will be to prove it for the modular group via an application of the argument principle and then extend it to the congruence subgroup Γ. In order to do this, we will need to show how to construct a modular form on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ given one on $\Gamma\backslash\mathbb{H}$. The following lemma provides the answer:

Lemma 2.9. *Let $f \in \mathcal{M}_k(\Gamma,\nu)$ and set*

$$F(z) = \prod_{\beta \in \Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} ((f|_k\beta)(z))^{n_\nu}.$$

Then $F \in \mathcal{M}_{kn_\nu d_\Gamma}(\mathrm{SL}_2(\mathbb{Z}))$.

Proof. If f is zero then so is F and the claim is obvious. So assume f is nonzero. First we must show that F is well-defined which is to say that it is independent of the representatives β. Indeed, if β and β' belong to the same orbit then $\beta'\beta^{-1} \in \Gamma$. Modularity and multiplicativity of the slash operator together with the fact that ν is of order n_ν show

$$((f|_k\beta')(z))^{n_\nu} = ((f|_k\beta)(z))^{n_\nu},$$

which proves F is well-defined. Since the product is finite, holomorphy of F is clear. To prove modularity, let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Because β runs over a complete set of representatives for $\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})$ so does $\beta\gamma$. Modularity then follows from a short computation again using multiplicativity of the slash operator. The growth condition, follows from that of f and the definition of F. This proves $F \in \mathcal{M}_{kn_\nu d_\Gamma}(\mathrm{SL}_2(\mathbb{Z}))$. $\square$

We can now state and prove the **valence formula**.

Theorem (Valence formula). *Suppose $f \in \mathcal{M}_k(\Gamma, \nu)$ is nonzero. Then*

$$\sum_{\mathfrak{a} \in \Gamma\backslash\hat{\mathbb{Q}}} v_{\mathfrak{a}}(f) + \sum_{z \in \mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}_\Gamma} \frac{2v_z(f)}{|\Gamma_z|} = \frac{kd_\Gamma}{12}.$$

Proof. We will first prove the valence formula for the modular group and then extend the result to Γ. Let $F \in \mathcal{M}_k(\mathrm{SL}_2(\mathbb{Z}))$ be nonzero so that its zeros are isolated. We claim that all of the zeros of F satisfy $\mathrm{Im}(z) < T$ for some positive T. To see this, let the Fourier series of F be given by

$$F(z) = \sum_{n \geq 0} a(n)e(nz).$$

Note that F is bounded as $z \to \infty$ since it is holomorphic at ∞. Therefore, under the change of variables $e(z) \mapsto q$ this means

$$G(q) = \sum_{n \geq 0} a(n)q^n,$$

converges absolutely provided $|q| < 1$ and where $G(0) = \lim_{z \to \infty} F(z)$. Then G is holomorphic on the interior of the unit disk and the order of vanishing of G at 0 is exactly $v_\infty(F)$. Therefore we may write

$$G(q) = q^{v_\infty(F)}H(q),$$

where H is holomorphic on the interior of the unit disk and $H(0) \neq 0$. Whence there exists a positive r such that $G(q)$ is nonzero for $0 < |q| < r$. But then $F(z)$ is nonzero for $e^{-2\pi ny} < r$ which is to say $y > -\frac{\log(r)}{2\pi y}$. Taking $T > -\frac{\log(r)}{2\pi y}$ proves the claim.

It follows that every zero ρ of F has ordinate less than T. Moreover, $\gamma\rho$ is a zero for every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ by modularity. For our choice of T, let $\eta = \sum_i \eta_i$ be the contour in Figure 2 where the horizontal line is along $t = T$ and the remaining contour is along the boundary of $\mathcal{F}$ except for indentations of radius ε around possible zeros of F along this boundary as well as the points i, ω, and $1+\omega$. Moreover, the paths are chosen to be symmetric with respect to the actions of S and T so that each zero ρ belonging to $\mathcal{F}$ has exactly one representative of its orbit occurring inside η except for when $\rho = i, \omega, 1+\omega$ which are excluded. As $v_z(F) = 0$ for every $z \in \mathcal{F} - \{i, \omega, 1+\omega\}$ that is not a zero of F, the argument principle gives

$$\frac{1}{2\pi i}\int_\eta \frac{F'}{F}(z)dz = \sideset{}{^*}\sum_{z \in \mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|},$$

where the $*$ indicates that we are excluding the orbits of i and ω. Now define integrals

$$I_i(\varepsilon) = \int_{\eta_i} \frac{F'}{F}(z)dz,$$

so that

$$\frac{1}{2\pi i}\int_\eta \frac{F'}{F}(z)dz = \sum_i I_i(\varepsilon).$$

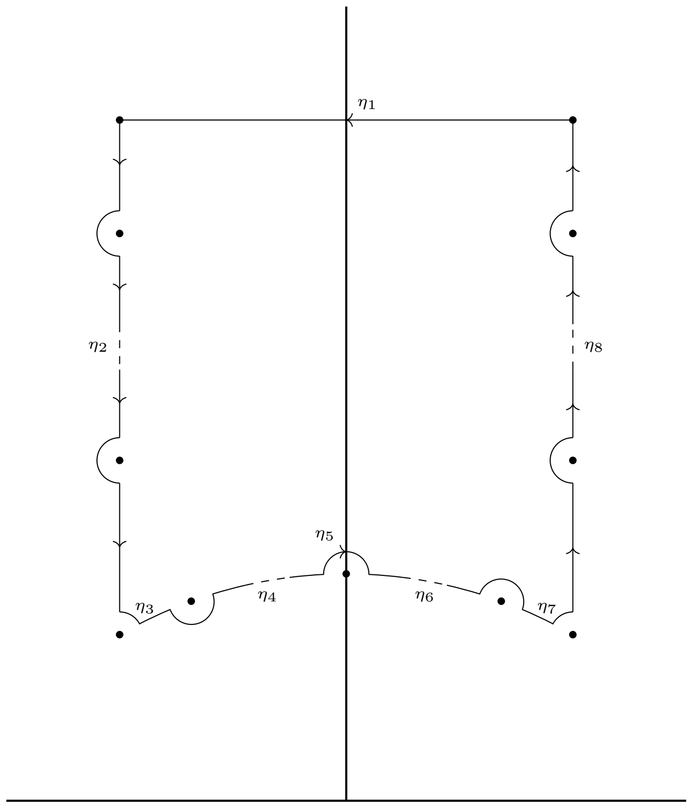

Figure 2: An indented standard fundamental domain contour.

We will compute the integrals $I_i(\varepsilon)$ and then take the limit as $\varepsilon \to 0$. For the integrals along η_2 and η_8, modularity implies $F(z+1) = F(z)$ and observe that $T^{-1}\eta_8 = -\eta_2$. Whence

$$I_2(\varepsilon) + I_8(\varepsilon) = 0,$$

so that these integrals cancel. For the integral along η_1, the change of variables $e(z) \mapsto q$ takes η_1 to the circle $|q| = e^{-2\pi T}$ transversed once with negative orientation and $F(z)$ is taken to $G(q)$. Then from our previous discussion of $G(q)$, we have

$$I_1(\varepsilon) = -v_\infty(F).$$

The integrals along η_3 and η_7, as well as η_5, are computed in a similar manner. Let θ_ε be the angle such that η_3 and η_7 end and begin at angles $\frac{\pi}{6} + \theta_\varepsilon$ and $\frac{5\pi}{6} - \theta_\varepsilon$ relative to ω and $1+\omega$ respectively. For η_3, write $F(z) = (z-\omega)^{v_\omega(F)} F_\omega(z)$ where $F_\omega(z)$ holomorphic and nonzero at ω. Then

$$I_3(\varepsilon) = \frac{1}{2\pi i} \int_{\eta_4} \left(\frac{v_\omega(F)}{z-\omega} + \frac{F'_\omega}{F_\omega}(z) \right) dz,$$

and the change of variables $z \mapsto \omega + \varepsilon e^{i\theta}$ shows

$$I_3(\varepsilon) = -\frac{\frac{\pi}{3} - \theta_\varepsilon}{2\pi} v_\omega(F) + O(\varepsilon).$$

Since $\theta_\varepsilon \to 0$ as $\varepsilon \to 0$, it follows that

$$\lim_{\varepsilon\to 0} I_3(\varepsilon) = -\frac{v_\omega(F)}{|\mathrm{SL}_2(\mathbb{Z})_\omega|},$$

as $|\mathrm{SL}_2(\mathbb{Z})_\omega| = 6$ by Proposition 1.10. We argue similarly for η_7. Write $F(z) = (z-(1+\omega))^{v_{1+\omega}(F)}F_{1+\omega}(z)$ where $F_{1+\omega}(z)$ holomorphic and nonzero at $1+\omega$. Then

$$I_7(\varepsilon) = \frac{1}{2\pi i}\int_{\eta_7}\left(\frac{v_{1+\omega}(F)}{z-(1+\omega)} + \frac{F'_{1+\omega}}{F_{1+\omega}}(z)\right) dz,$$

and the change of variables $z \mapsto (1+\omega)+\varepsilon e^{i\theta}$ shows

$$I_7(\varepsilon) = \frac{\frac{\pi}{3}-\theta_\varepsilon}{2\pi}v_{1+\omega}(F) + O(\varepsilon).$$

Whence

$$\lim_{\varepsilon\to 0} I_7(\varepsilon) = -\frac{v_{1+\omega}(F)}{|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|},$$

where $|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}| = 6$ by Proposition 1.10. For the integral along η_5, write $F(z) = (z-i)^{v_i(F)}F_i(z)$ where $F_i(z)$ is holomorphic and nonzero at i. Then

$$I_5(\varepsilon) = \frac{1}{2\pi i}\int_{\eta_5}\left(\frac{v_i(F)}{z-(i)} + \frac{F'_i}{F_i}(z)\right) dz,$$

and the change of variables $z \mapsto i+\varepsilon e^{i\theta}$ gives

$$I_5(\varepsilon) = -\frac{v_i(F)}{2} + O(\varepsilon).$$

Therefore

$$\lim_{\varepsilon\to 0} I_5(\varepsilon) = -\frac{2v_i(F)}{|\mathrm{SL}_2(\mathbb{Z})_i|},$$

as $|\mathrm{SL}_2(\mathbb{Z})_i| = 4$ by Proposition 1.10. For the integrals along η_4 and η_6, let C_ε be the finite union of segments along η_4 that lie on the boundary of $\mathcal{F}$. Modularity implies $F\left(-\frac{1}{z}\right) = z^kF(z)$ and notice that $S^{-1}\eta_6 = -\eta_4$. Hence

$$I_4(\varepsilon) + I_6(\varepsilon) = -\frac{1}{2\pi i}\int_{C_\varepsilon}\frac{k}{z}\,dz.$$

As $\varepsilon \to 0$, C_ε tends to the arc on the unit circle between radii $\frac{\pi}{2}$ and $\frac{2\pi}{3}$ with negative orientation. From the change of variables $z \mapsto e^{i\theta}$, we find that

$$\lim_{\varepsilon\to 0}\left(I_4(\varepsilon)+I_6(\varepsilon)\right) = \frac{k}{12}.$$

Altogether, this means

$$\lim_{\varepsilon\to 0}\frac{1}{2\pi i}\int_\eta \frac{F'}{F}(z)dz = \frac{k}{12} - v_\infty(F) - \frac{v_\omega(F)}{|\mathrm{SL}_2(\mathbb{Z})_\omega|} - \frac{v_{1+\omega}(F)}{|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|} - \frac{2v_i(F)}{|\mathrm{SL}_2(\mathbb{Z})_i|}, \tag{7}$$

whence

$$v_\infty(F) + \frac{v_\omega(F)}{|\mathrm{SL}_2(\mathbb{Z})_\omega|} + \frac{v_{1+\omega}(F)}{|\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|} + \frac{2v_i(F)}{|\mathrm{SL}_2(\mathbb{Z})_i|} + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}}^{*} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \frac{k}{12}.$$

Since ω and $1+\omega$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent, $v_\omega(F) = v_{1+\omega}(F)$ and $|\mathrm{SL}_2(\mathbb{Z})_\omega| = |\mathrm{SL}_2(\mathbb{Z})_{1+\omega}|$ by Proposition 1.10. So the previous identity becomes

$$v_\infty(F) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \frac{k}{12},$$

which is precisely the claim for the modular group. It remains to prove the claim for Γ. So let $f \in \mathcal{M}_k(\Gamma,\nu)$ be nonzero. By Lemma 2.9 we may take

$$F(z) = \prod_{\beta\in\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} ((f|_k\beta)(z))^{n_\nu},$$

and obtain

$$v_\infty(F) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \frac{kmd_\Gamma}{12}.$$

Observe that the Fourier series of F is a product of the Fourier series for f over a complete set of Γ-inequivalent cusps each with multiplicity m. Since the Fourier series of f at the $\mathfrak{a}$ cusp is $h_\mathfrak{a}$-periodic, the index of the first nonzero term in the Fourier series of F is

$$v_\infty(F) = \sum_{\mathfrak{a}\in\Gamma\backslash\hat{\mathbb{Q}}} \frac{mv_\mathfrak{a}(f)}{h_\mathfrak{a}}.$$

Lastly, modularity implies $F(z) = 0$ if and only if $f(\beta z) = 0$ for some β in which case the zero is of multiplicity m. From this and the orbit-stabilizer theorem, we find that

$$\frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \sum_{\beta\in\Gamma\backslash\mathrm{SL}_2(\mathbb{Z})} \frac{2mv_{\beta z}(f)}{|\Gamma_{\beta z}|}.$$

But then Proposition 1.9 further implies

$$\sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}} \frac{2v_z(F)}{|\mathrm{SL}_2(\mathbb{Z})_z|} = \sum_{z\in\Gamma\backslash\mathcal{F}_\Gamma} \frac{2mv_z(f)}{|\Gamma_z|}.$$

Substituting these identities into the formula for the modular group gives

$$\sum_{\mathfrak{a}\in\Gamma\backslash\hat{\mathbb{Q}}} v_\mathfrak{a}(f) + \sum_{z\in\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{F}_\Gamma} \frac{2v_z(f)}{|\Gamma_z|} = \frac{kd_\Gamma}{12},$$

as desired. $\square$

We can now prove that $\mathcal{M}_k(\Gamma,\nu)$ and $\mathcal{S}_k(\Gamma,\nu)$ are finite dimensional.

Theorem 2.10. *$\mathcal{M}_k(\Gamma,\nu)$ is finite dimensional.*

Proof. Let $f \in \mathcal{M}_k(\Gamma,\nu)$. Recall that $v_{\mathfrak{a}}(f)$ and $v_z(f)$ are nonnegative. In particular, $v_\infty(f)$ is nonnegative and the valence formula implies $v_\infty(f) \le \frac{kd_\Gamma}{12}$ whenever f is nonzero. Let n be a positive integer such that $n \ge \frac{kd_\Gamma}{12}+1$ and consider the linear map $F : \mathcal{M}_k(\Gamma,\nu) \to \mathbb{C}^n$ sending f to its first n Fourier coefficients. Then F is injective since $v_\infty(f) < n$ and hence $\mathcal{M}_k(\Gamma,\nu)$ is finite dimensional. $\square$

For further developments, we will require the Petersson inner product. For $f,g \in \mathcal{S}_k(\Gamma,\nu)$ we define their **Petersson inner product** $\langle f,g\rangle_\Gamma$ by

$$\langle f,g\rangle_\Gamma = \frac{1}{V_\Gamma}\int_{\Gamma\backslash\mathbb{H}} f(z)\overline{g(z)}\mathrm{Im}(z)^k\,d\mu.$$

If the congruence subgroup Γ is clear from context we will suppress the dependence. The integral is well-defined because the integrand is Γ-invariant. Moreover, as f and g have exponential decay at the cusps, the integral is also locally absolutely uniformly convergent. We also extend the Petersson inner product to any f or g provided the integrand is Γ-invariant and the integral is locally absolutely uniformly convergent.

The Petersson inner product makes $\mathcal{S}_k(\Gamma,\nu)$ into a complex Hilbert space. Indeed, the Petersson inner product is clearly linear on $\mathcal{S}_k(\Gamma,\nu)$. It is also positive definite since

$$\langle f,f\rangle = \frac{1}{V_\Gamma}\int_{\mathcal{F}_\Gamma} |f(z)|^2\mathrm{Im}(z)^k\,d\mu,$$

which is clearly nonnegative and zero if and only if f is identically zero. A short computation together using the fact that $\overline{d\mu} = d\mu$, which holds since $d\mu = \frac{dx\,dy}{y^2}$, proves conjugate symmetry. Whence $\mathcal{S}_k(\Gamma,\nu)$ is a complex inner product space with respect to the Petersson inner product. Since $\mathcal{S}_k(\Gamma,\nu)$ is finite dimensional by Theorem 2.10, it follows immediately that $\mathcal{S}_k(\Gamma,\nu)$ is a complex Hilbert space.

We will now take $\nu = \chi$ for χ a Dirichlet character modulo N. Our goal will be to show that $\mathcal{M}_k(\Gamma,\chi)$ decomposes as

$$\mathcal{M}_k(\Gamma,\chi) = \mathcal{S}_k(\Gamma,\chi) \oplus \mathcal{E}_k(\Gamma,\chi),$$

where $\mathcal{S}_k(\Gamma,\chi)$ is spanned by the Poincaré series of positive index. Let $\mathfrak{a}$ be a cusp of $\Gamma\backslash\mathbb{H}$ and choose a scaling matrix $\sigma_{\mathfrak{a}}$. Suppose $f \in \mathcal{S}_k(\Gamma,\nu)$ and let its Fourier series at the $\mathfrak{a}$ cusp be

$$(f|_k\sigma_{\mathfrak{a}})(z) = \sum_{n\ge1} a_{\mathfrak{a}}(n)e\left(\frac{nz}{h_{\mathfrak{a}}}\right).$$

For any positive integer m, define linear functionals $\phi_{m,k,\chi,\mathfrak{a}} : \mathcal{S}_k(\Gamma,\chi) \to \mathbb{C}$ by

$$\phi_{m,k,\chi,\mathfrak{a}}(f) = a_{\mathfrak{a}}(m).$$

Since $\mathcal{S}_k(\Gamma,\chi)$ is a finite dimensional complex Hilbert space, the Riesz representation theorem implies that there exists unique $v_{m,k,\chi,\mathfrak{a}} \in \mathcal{S}_k(\Gamma,\chi)$ such that

$$\langle f, v_{m,k,\chi,\mathfrak{a}}\rangle = a_{\mathfrak{a}}(m).$$

We would like to know what these cusp forms are. It turns out that $v_{m,k,\chi,\mathfrak{a}}(z)$ will be the Poincaré series $P_{m,k,\chi,\mathfrak{a}}(z)$ up to a constant.

Theorem 2.11. *Let $f \in \mathcal{S}_k(\Gamma, \chi)$ have Fourier coefficients $a_{\mathfrak{a}}(n)$ at the $\mathfrak{a}$ cusp of $\Gamma\backslash\mathbb{H}$. Then*

$$\langle f, P_{m,k,\chi,\mathfrak{a}}\rangle = \frac{\Gamma(k-1)h_{\mathfrak{a}}^k}{V_\Gamma(4\pi m)^{k-1}} a_{\mathfrak{a}}(m),$$

for all positive integers m.

Proof. We will compute

$$\langle f, P_{m,k,\chi,\mathfrak{a}}\rangle = \frac{1}{V_\Gamma}\int_{\mathcal{F}_\Gamma} f(z)\overline{P_{m,k,\chi,\mathfrak{a}}(z)}\mathrm{Im}(z)^k\, d\mu.$$

A short computation using Equation (2), modularity, and the cocycle condition, shows that this integral is

$$\frac{1}{V_\Gamma}\int_{\mathcal{F}_\Gamma}\sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} j(\sigma_{\mathfrak{a}}, \sigma_{\mathfrak{a}}^{-1}\gamma z)^{-k} f(\gamma z) e\left(-\frac{m\overline{\sigma_{\mathfrak{a}}^{-1}\gamma z}}{h_{\mathfrak{a}}}\right)\mathrm{Im}(\sigma_{\mathfrak{a}}^{-1}\gamma z)^k\, d\mu.$$

Applying the change of variables $z \mapsto \sigma_{\mathfrak{a}} z$ gives

$$\frac{1}{V_\Gamma}\int_{\mathcal{F}_{\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}}}\sum_{\gamma\in\Gamma_{\mathfrak{a}}\backslash\Gamma} j(\sigma_{\mathfrak{a}}, \sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{a}} z)^{-k} f(\gamma\sigma_{\mathfrak{a}} z) e\left(-\frac{m\overline{\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{a}} z}}{h_{\mathfrak{a}}}\right)\mathrm{Im}(\sigma_{\mathfrak{a}}^{-1}\gamma\sigma_{\mathfrak{a}} z)^k\, d\mu.$$

Making the further change of variables $\gamma \mapsto \sigma_{\mathfrak{a}}\gamma\sigma_{\mathfrak{a}}^{-1}$ results in

$$\frac{1}{V_\Gamma}\int_{\mathcal{F}_{\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}}}\sum_{\gamma\in\Gamma_\infty^{(h_{\mathfrak{a}})}\backslash\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}} (f|_k\sigma_{\mathfrak{a}})(\gamma z) e\left(-\frac{m\overline{\gamma z}}{h_{\mathfrak{a}}}\right)\mathrm{Im}(\gamma z)^k.$$

Now unfold to obtain

$$\frac{1}{V_\Gamma}\int_{\Gamma_\infty^{(h_{\mathfrak{a}})}\backslash\mathbb{H}} (f|_k\sigma_{\mathfrak{a}})(z) e\left(-\frac{m\overline{z}}{h_{\mathfrak{a}}}\right)\mathrm{Im}(z)^k\, d\mu.$$

Substituting in the Fourier series of f at the $\mathfrak{a}$ cusp and interchanging the inner sum and integral by absolute convergence, a short computation gives

$$\frac{h_{\mathfrak{a}}}{V_\Gamma}\int_0^\infty a_{\mathfrak{a}}(m) e^{-\frac{4\pi m y}{h_{\mathfrak{a}}}} y^k\, \frac{dy}{y^2},$$

in view of Equation (6) because every term integrates to zero except the m-th term. Performing the change of variables $y \mapsto \frac{h_{\mathfrak{a}} y}{4\pi m}$, this latter integral is recognized as the Gamma function and we obtain

$$\frac{\Gamma(k-1)h_{\mathfrak{a}}^k}{V_\Gamma(4\pi m)^{k-1}} a_{\mathfrak{a}}(m),$$

completing the proof. □

It follows immediately from this result that

$$v_{m,k,\chi,\mathfrak{a}}(z) = \frac{V_\Gamma (4\pi m)^{k-1}}{\Gamma(k-1) h_{\mathfrak{a}}^k} P_{m,k,\chi,\mathfrak{a}}(z).$$

Thus the Poincaré series $P_{m,k,\chi,\mathfrak{a}}(z)$ of positive index extract the Fourier coefficients of f at the $\mathfrak{a}$ cusp up to constants. With this result in hand we can prove that these series span the space of cusp forms and that the orthogonal complement in the .

Theorem 2.12. *The Poincaré series $P_{m,k,\chi}(z)$ of positive index span $\mathcal{S}_k(\Gamma, \chi)$.*

Proof. Let $f \in \mathcal{S}_k(\Gamma, \chi)$ have Fourier coefficients $a(n)$. Since $\Gamma(k-1)$ is nonzero, Theorem 2.11 implies $\langle f, P_{m,k,\chi,\mathfrak{a}} \rangle = 0$ if and only if $a(m) = 0$. So f is orthogonal to all of the Poincaré series $P_{m,k,\chi}$ of positive index if and only if every Fourier coefficient $a(m)$ is zero. But this happens if and only if f is identically zero. $\square$